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Designing Size Inclusive Fashion Assortments

Alireza Alavi

Desautels Faculty of Management, McGill University, Montreal, Quebec H3A 1G5, alireza.alavi@mail.mcgill.ca

Mehmet Gumus

Desautels Faculty of Management, McGill University, Montreal, Quebec H3A 1G5, mehmet.gumus@mcgill.ca

Javad Nasiry

Desautels Faculty of Management, McGill University, Montreal, Quebec H3A 1G5, javad.nasiry@mcgill.ca

Abstract. Problem Definition: Assortment optimization methods help fashion retailers improve profitability and customer satisfaction. By focusing primarily on revenue maximization, these methods often overlook size inclusivity, leading to the under-representation of plus-sized customers. This raises fairness concerns and can negatively impact brand perception. **Methodology:** We develop a choice model that allows for size substitution and introduce a measure of pairwise fairness among customer size groups. We then integrate the two into an online fashion retailer’s in-season assortment optimization problem in which the retailer can randomize its display over multiple assortments. By using *selection-based* formulation, we transform the combinatorial assortment problem into a linear program over option-selection probabilities, and develop a backward mapping that recovers an optimal randomized assortment policy from the LP solution. We prove that the two problems are equivalent, so the LP delivers an exact (rather than approximate) optimum while remaining polynomial in the number of products and size groups. Building on this framework, we study several behavioral and design extensions—including option-substitution—and derive the corresponding optimal policies from closely related LPs. We estimate the size substitution coefficients, calibrate our model using a dataset provided by our industrial partner, and run extensive numerical experiments. **Managerial implications:** Numerical experiments show that our method can reduce unfairness by 30% with only a 1.5% drop in revenue. They further reveal that the drop in unfairness happens not by eradicating unavailability for all size groups, but by redistributing it more evenly across them. Finally, allowing customers to filter by size before browsing, or to revisit the site following a size stockout, expands the revenue–fairness frontier, so that the retailer can simultaneously raise revenue and lower unfairness relative to the baseline.

Key words: assortment optimization, size inclusivity, choice model, pairwise fairness, randomization

1. Introduction

Fashion retailers utilize assortment optimization methods to enhance profitability, minimize excess inventory, and boost customer satisfaction (*Style Arcade* 2024). Despite their financial successes, these methods fail to address the *size inclusivity* issue, and plus-sized people experience significantly higher unavailability rates. For example, an analysis of the top ten womenswear retailers in the UK by *Retail-Week* (2022) revealed that the availability rate for size 20 is just 53%, compared to 76% for size 10. This disparity in size availability across different colors and styles is a “subtle form of social discrimination,” and it negatively impacts customers’ brand perception (Volonté 2021). Knowing that the perception of a brand’s diversity

and inclusivity efforts affects the purchase decision of 80% of customers (Kantar 2025), the absence of size inclusivity seems more surprising.

Size inclusivity is not a recent challenge, but a legacy of decades-old sizing standards (*The Interline* 2025), and it continues to be an industry-wide issue today (*Business Insider* 2025). For instance, while 67% of American women wear size 14 or above (plus-sized), most options are not offered in these sizes (*Time* 2016). The roots of non-inclusive fashion retailing can be traced back to design complexities and higher production costs (*Metro* 2019). Moreover, supply-side constraints such as *pre-pack ordering*—where vendors ship garments only in bundles containing preset size ratios—and mandatory minimum order quantities (*Vogue* 2022) together pose upstream limitations that a single retailer cannot easily override. Although these constraints explain (and, from an operational perspective, help to justify) the unavailability of certain sizes, the retailer still controls a critical lever that shapes customer judgments: the assortment design.

Because 73% of customers never visit the second page of a category (*Avatria* 2021) and online fashion retailers, on average, do not know the sizes of more than half of arriving customers (*Bluecore* 2025), we develop a model to answer the following question: *How can online fashion retailers optimize the first-page assortment to maximize revenue while mitigating concerns about unfair size representation?* We then extend the model to settings in which customers reveal their sizes before browsing.

Our study is driven by a collaboration with one of Europe’s largest online fashion retailers. Like many in the industry, a challenge for our partner is to introduce a wide range of new products to meet rapidly evolving customer preferences. During a typical spring/summer season, the retailer offers around 30,000 options in different categories. Table 1 shows the percentage of customers and the mid-season unavailability rate for different sizes in the women’s shirts category in a data sample from our industry partner. Although anecdotal and based on censored sales data, Table 1 provides confirmatory evidence on the *systematic discrimination* that our work aims to alleviate: while size groups XXL and L have the same number of customers, the unavailability rate for size group XXL is more than twice that of size group L.

An informative observation comes from a shift in our industry partner’s strategy. Initially, the retailer largely excluded size XXL from its production line, so only 0.2% of the SKUs introduced into the online inventory for women’s shirts were size XXL. After the retailer changed its strategy, this share increased to 28%. Yet, as Table 1 shows, plus-sized customers still experience substantially higher levels of size unavailability, highlighting the importance of incorporating size inclusivity into mid-season decisions.

Size	XS	S	M	L	XL	XXL
Unavailability Rate	22.2%	19.5%	20.4%	29.8%	32.2%	68.7%
Customer Distribution	10%	20%	22%	16%	16%	16%

Table 1 Customer size distribution and mid-season size unavailability rates in the women’s shirts category.

Using this as our main setting, we assume inventory availability is exogenous and study the in-season assortment optimization problem of maximizing revenue while reducing unfair SKU representation

across the customer base. Most SKUs managed by fast-fashion retailers—more than 75% at our partner retailer—fit this setting: once the initial shipment reaches the central warehouse for the online store, no replenishment occurs from suppliers. Thus, after the season begins, availability is effectively fixed, and the retailer’s role shifts from pre-season assortment decisions (often called category planning) to in-season assortment choices that maximize sales probability. We modify this problem by incorporating a fairness notion based on size availability. To the best of our knowledge, this is the first paper in the assortment optimization literature to make size inclusivity its core objective.

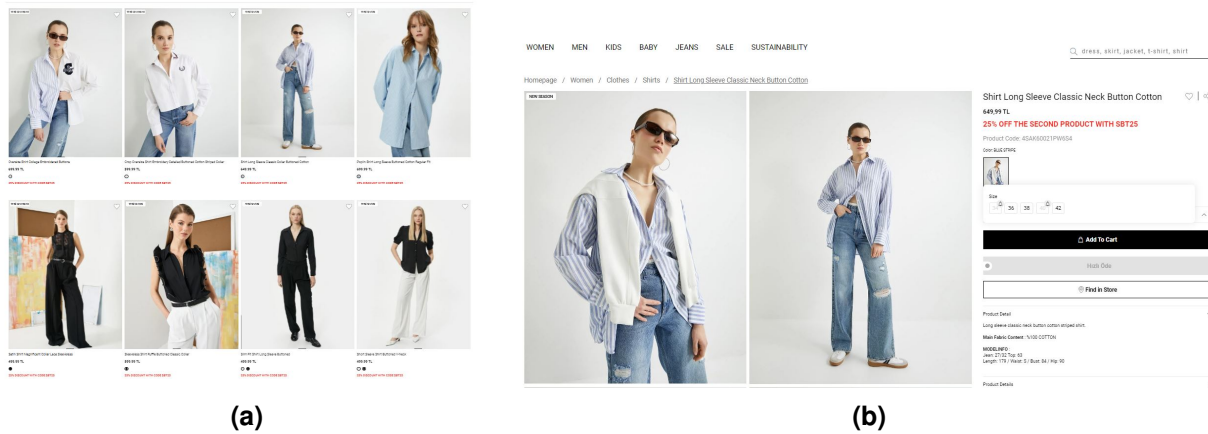


Figure 1 Figure (a): The category page for women’s shirts. Figure (b): A sample option’s page.

To explain our main contributions, we review the problem setup and terminology. An option is a style and color combination, and an SKU is a particular size of that option. The sale of an SKU from option j brings the revenue r_j to the retailer. The inventory has n distinct options, and the retailer decides the option assortment $S \subseteq \{1, \dots, n\}$. An option may not be available in all sizes, so we denote the size configuration of option j by $C_j \subseteq G$ where $G = \{1, \dots, g\}$ is the set of SKU sizes. We let $\mathcal{C} := \{C_j | j \in S\}$.

In the baseline model, a customer first visits a product category page similar to Figure 1a, considers all options in that category, and chooses option j according to a Multinomial Logit (MNL) model. After clicking on option j , she lands on an option page similar to Figure 1b. After observing C_j , she may choose an SKU from option j or leave the store. If her preferred size is unavailable for option j , she may substitute it with another size of the same option. We categorize customers into g size groups based on the SKU size that fits them best and define ‘unavailability’ as the probability that a customer in a given size group experiences size unavailability. This allows us to develop a *pairwise fairness* metric that measures the gap between the highest and lowest unavailability levels across size groups. (Notably, our methods also apply to *max-min (Rawlsian) fairness* metrics.) We set the objective as expected revenue minus the fairness metric multiplied by λ , where λ represents the *price of unfairness*. We next summarize our main results and contributions.

First, we prove that when all sizes are available for all options, the optimal option assortment has the same revenue-ordered solution developed in Talluri and Van Ryzin (2004) and, as such, is unaffected by fairness concerns. However, full size availability is not realistic, and retailers offer a *broken assortment* with limited size availability for the majority of the selling season. Displaying such options potentially causes fairness concerns. In such cases, the optimal solution depends on λ . Being an online retailer, the retailer has the ability to randomize among multiple assortments instead of displaying only one assortment (Wang et al. 2024). With that, we develop the *profitable yet size-inclusive* static assortment problem.

Second, we employ and extend the *selection-based* method to solve the problem exactly. Because there are 2^n feasible assortments, the retailer's problem has exponentially many decision variables. We therefore first transform it into a linear program (LP) in which, instead of choosing a display probability for each assortment, the retailer chooses a display probability for each option. This LP has $\mathcal{O}(n)$ decision variables and constraints, allowing it to be solved quickly by commercial solvers. We then transform the optimal LP solution back into assortment display probabilities. In addition to being optimal, this solution uses at most g^2 distinct assortments, which supports its practical implementation.

Third, we extend the problem and our solution methods to characterize the optimal solutions under several consumer behaviors. First, we consider the case in which a customer applies a size filter on the category page before choosing an option. Second, we extend the baseline model to allow for option substitution, whereby customers may revisit the category page and choose another option when their preferred size is unavailable. Third, we combine option substitution with size filtering. Fourth, we allow for endogenous configuration decisions, whereby the retailer chooses both the options to display and the configurations in which they are displayed. Beyond establishing existence and tractability, these extensions yield two managerial insights. Encouraging customers to reveal their size first (e.g., through default size filters) allows the retailer to tailor assortments and weakly improve both revenue and fairness. Conversely, anticipating that customers may revisit the site after encountering size unavailability creates an incentive for the retailer to display more revenue-prioritizing (and thus less fair) first-stage assortments and rely on the second stage to mitigate unavailability—so revisits do not always reduce overall unfairness.

Finally, we estimate the size substitution coefficient using a granular dataset provided by our industry partner and the difference-in-differences method of Li et al. (2023). The treatment group consists of SKUs whose neighboring size is unavailable, while the control group consists of SKUs from the same option that are unaffected by neighbor unavailability. The regression results imply an approximate size substitution coefficient of 25%, close to the estimate in Li et al. (2023). The estimated value is economically meaningful: a quarter of customers in a given size group are willing to purchase a neighboring size when their preferred size is missing, which makes the supply of substitutable sizes a real lever in the retailer's hands. We use this estimate to calibrate our choice model and run simulations to evaluate the revenue-unfairness trade-off. The simulations yield two key managerial insights. First, the retailer can reduce unfairness by nearly 30%

with only a 1.5% drop in revenue, so size inclusivity comes at a modest revenue cost. Second, fairness-oriented assortments do not eliminate unavailability for plus-sized groups; instead, they redistribute it more evenly across all size groups by reallocating size coverage across the displayed options—and the width of the optimal assortment may either expand or shrink as the retailer places more weight on fairness.

The rest of the paper is organized as follows. We review the relevant literature in Section 2. We develop the choice model, the fairness metric, and the objective function in Section 3. The structure of the optimal solution and the solution method are discussed in Section 4. Section 5 extends the baseline model to more general choice behaviors. Section 6 presents the managerial insights. Finally, we discuss possible extensions and conclude the paper in Section 7.

2. Literature Review

Our work relates to the literature on assortment optimization and fair operations. In this section, we review the relevant papers from each area of literature.

2.1. Choice Models and Assortment Optimization

The Multinomial Logit (MNL) model, developed by McFadden (1972), is widely used to model choice behavior due to its tractability. The model however has important limitations such as ignoring customer heterogeneity (Aouad et al. 2019) and exhibiting the ‘Independence of Irrelevant Alternatives’ (IIA) property (Blanchet et al. 2016). Among alternative choice models that aim to address these issues are Mixed-Multinomial Logit (MMNL) model (McFadden and Train 2000) where customer valuations can be heterogeneous, and the consider-then-choose model (Aouad et al. 2021) where customers consider a subset of products before ranking them. We refer to Feng et al. (2022) for a comprehensive literature review. We use the MNL choice model to conceptualize how customers choose among the options, and further add a layer to capture how customers choose among the sizes available for an option. As such, our research is also related to sequential search in discrete choice, recently studied by Kosilova and Alptekinoglu (2025).

Talluri and Van Ryzin (2004) prove that, under an MNL choice model, the optimal assortment has a revenue-ordered form. However, finding the optimal assortment with multiple customer groups based on the MMNL model is NP-complete (Rusmevichientong et al. 2014). Some studies attempt to tackle this issue by finding quick, efficient heuristics (Feldman and Topaloglu 2015) or providing polynomial-time algorithms with an approximation guarantee (Désir et al. 2022). We study a case where customers in different size groups have the same valuation for an option, so we do not have this complexity issue.

Inventory stockouts are among the main factors that affect customer choice behavior. Recently, there has been an increase in research aimed at curtailing the effects of stockouts. For instance, Gopalakrishnan et al. (2025) investigates the extent to which cross-store fulfillment can recover sales lost due to stockouts. Nonetheless, it is important to know how stockouts affect customer behavior. In the face of stockouts, customers may opt to substitute another product. This ‘stockout-based substitution’ behavior (Kök et al.

2015) has been studied in the assortment optimization literature (Honhon et al. 2010) and in the demand estimation literature (Musalem et al. 2010). Stockouts in fashion retailing are more nuanced as they can affect both options and SKUs. A stockout at the SKU level leads to a broken assortment where some sizes of an option are out of stock (Caro and Gallien 2010). The estimations of Boada-Collado and Martínez-de Albéniz (2020) demonstrate that a broken assortment has a strong negative impact on sales. Li et al. (2023) study how customers react to broken assortments by analyzing a granular dataset from a shoe store chain in China. They estimate that size stockouts result in size substitution (that is, substituting with neighboring sizes) in almost 50% of the cases while only 20% of the cases lead to option substitution. Our research aims to integrate the size substitution behavior into the customer choice model and characterize the optimal assortment for a fashion retailer.

A paper close to ours is Akchen and Caro (2026). They develop a choice model that accounts for the size substitution effect. They then estimate the size substitution coefficient in their model using a real dataset and solve the assortment and inventory optimization problem. Our research differs in several aspects. First, they assume customers place lower valuations on broken assortments, but we assume they assign lower fit probability to them. This approach allows us to develop a closed-form solution with and without a broken assortment. Second, they model a dynamic setting in which assortment and inventory can be optimized jointly, whereas we assume a static setting in which inventory is set exogenously. Lastly, we take into account size inclusivity and fairness considerations so that different size groups do not experience systematic discrimination, i.e., considerably different levels of size unavailability.

Methodologically, we employ and adjust the *selection-based* approach. The problem of choosing the right assortment(s) is of a combinatorial nature. To address this issue, the authors of Gallego et al. (2015) developed an approach that, instead of focusing on which assortment to display, determines the *sales quantity* of each item. They then develop a duality-based algorithm to transform the outcomes to the original assortment-based format. In parallel, Topaloglu (2013) used a similar approach for joint assortment and inventory decisions. Fundamentally a decomposition method, this approach vastly simplifies the problem and is directly applicable to stochastic revenue management. Nonetheless, because it may suggest solutions that involve more than one assortment (e.g., displaying assortment A for the first half of a season and assortment B for the second half), it cannot be readily applied to the static assortment optimization setting. However, with the idea of *randomization* gaining traction in recent years (see, for instance, Wang et al. (2024) and Zhu et al. (2025)), this approach has found its way to static settings. Closest to our setup, Zhu et al. (2025) model the assortment optimization problem with market share constraints, where the market share of a certain class of products must be either zero or within a certain range. After proving the complexity of their problems, they develop approximation schemes for both the randomized and deterministic cases. Our research differs from theirs in the choice model we utilize, the nature of the pairwise fairness assumptions we imposed, and the extensions with tractable optimal solutions we propose.

2.2. Fair Operations

Two key studies by Bertsimas et al. (2011, 2012) propose ‘the price of fairness’ as a relative measure to capture the trade-off between fairness and efficiency in a resource allocation problem. Such trade-offs are studied in a variety of contexts, varying from air traffic flow management (Barnhart et al. 2012), to kidney exchange (Bertsimas et al. 2013), to portfolio management (Iancu and Trichakis 2014), to gender inequality in restroom access (Farajollahzadeh and Hu 2021). To review the literature on fair and equitable practices in the operations literature, we refer to Karsu and Morton (2015).

An approach to incorporate fairness considerations into a decision maker’s problem is to set a weighted average of the firm’s revenue and a notion of social responsibility as the objective function for the decision maker. For instance, Wu et al. (2020) consider a combination of revenue and investments in socially responsible activities as the objective function. Hu et al. (2021) integrate customer surplus into a news vendor’s profit function. Sumida et al. (2021) add customers’ utility to the objective function in an assortment optimization problem. Recent work proposes unified frameworks for convex fairness measures that enable systematic incorporation into optimization problems Tsang and Shehadeh (2025). The objective function in our paper is a weighted average of the retailer’s revenue and a pairwise fairness metric comparing the shopping experience of different size groups.

There has been a recent interest in incorporating fairness in assortment optimization. Levi et al. (2024) propose a rigorous definition of inclusivity in the product line design and assortment optimization. Chen et al. (2022) and Lu et al. (2023) solve a static assortment optimization problem with fairness constraints on the outcome for different products. Barré et al. (2023) follow the same goal but in a dynamic setting. While the pairwise fairness we use is inspired by Chen et al. (2022), and our ‘randomization’ approach is similar to that in Lu et al. (2023), we are concerned with fairness among different customer groups in this research.

3. Problem Definition

In this section, we first define the fundamentals in our model. We then outline the choice model and the firm’s assortment optimization problem subject to fairness considerations.

Retailers usually categorize similar products together, e.g., women’s T-shirts. In each category, we assume the option-SKU hierarchy we introduced in Section 1. The sale of an SKU from option j brings revenue r_j to the retailer. We index options in descending revenue order: $r_1 \geq r_2 \geq \dots \geq r_n$. We denote the *sizing system* with G , which represents the possible SKU sizes, and define $g := |G|$.

We now elaborate on the notion of *configuration*. Consider an option j which the retailer offers in sizes $G = \{XS, S, M, ML, L, XL, XXL\}$. However, not all sizes of an option are available all the time. So the configuration of option j , $C_j \subseteq G$, denotes the sizes in which option j is available in the inventory. For instance, if $C_j = \{S, M, XXL\}$, then option j is available in small, medium, and double extra-large sizes. Here C_j does not represent how many of each SKU is available in the inventory; rather, $M \in C_j$

means at least one SKU of option j with size medium is available in the inventory. If there is an option $j \in N := \{1, 2, \dots, n\}$ for which we have $C_j \neq G$, we say option j has a ‘broken assortment’.

An assortment $S \subseteq N$ is the set of options that the retailer displays on its website. When the retailer includes option j in the assortment, all SKUs in C_j are displayed to the customer. (We relax this assumption in Section 5.4.) Online retailers, including our partner retailer, typically display all available inventory options across multiple pages on their websites. Nonetheless, when searching, customers rarely go beyond the initial pages. For instance, 73% of customers never visit the second page of a category (Avatria 2021). So, it is critical to determine which options to display on the first page. In that case, S represents the assortment of options on the first page. We summarize the core notions in our model in Definition 1.

DEFINITION 1. A configuration for option j , i.e., C_j , is the set of available sizes of that option in the inventory. An assortment S represents the set of distinct options displayed by the retailer. Finally, the broken assortment issue exists when $C_j \neq G$ for at least one $j \in N$.

3.1. The Choice Model

We develop a two-stage choice model in which customers first choose an option, then select the size to purchase. In Section 5.1, we consider a choice model in which customers first filter by size and then select among the remaining options. While our choice model does not follow a consider-then-choose structure, it draws on the literature supporting that structure. This literature assumes customers process information sequentially due to limitations in simultaneous consideration; see, e.g., Gilbride and Allenby (2004).

We denote a customer’s physical size with a one-dimensional variable $b \in \mathbb{R}^+$. Each customer has a size substitution tolerance t . (The threshold may be asymmetric with respect to upper and lower size substitutions.) If her distance to SKU size i (i.e., $|b - i|$) is shorter than t , she picks it; if there is more than one SKU closer than t , she picks the closest one. In general, this threshold and sensitivity to the distance can vary for larger and smaller sizes. We also define ‘size group i ’ as the group of customers who, when facing a full configuration (i.e., $C_j = G$), pick SKU size i . We denote the fraction of customers in size group i with ξ^i , such that $\sum_{i \in G} \xi^i = 1$.

In a typical online apparel shopping experience, the customer chooses the apparel category first and lands on the category page showcased in Figure 1a, where the first stage begins. The category page of online apparel retailers is often ‘size-agnostic,’ presenting all available options regardless of their configuration. We assume that the customer observes the option assortment S and, following an MNL choice model, chooses the option with the highest utility. In an MNL model, a customer’s utility for option j is $U_{j,\text{customer}} = u_j + \varepsilon_{j,\text{customer}}$, where $\varepsilon_{j,\text{customer}}$ ’s are i.i.d. samples of the standard Gumbel distribution. There is also a no-purchase option with u_0 . Assuming a customer chooses the option that maximizes her utility, the probability of choosing option j is:

$$\Pr(j|S) = \frac{v_j}{v_0 + \sum_{m \in S} v_m}, \quad (1)$$

where $v_j = \exp(u_j)$ is the preference weight for option j . We assume v_j is independent of q , as we do not believe physical size affects customer preferences over options. A similar assumption is applied by Li et al. (2023) and Akchen and Caro (2026).

After choosing option j , the customer lands on option j 's webpage (see Figure 1b for an example), and the second stage begins. Upon observing C_j , the customer evaluates her distance to the available sizes. She then chooses the size with the lowest distance. If none of the available sizes are closer than t , the customer leaves the store empty-handed. In Section 5.2, we consider the possibility of 'option substitution,' whereby customers whose desired size is unavailable may continue searching for other options rather than leaving the store immediately.

Because leaving the store empty-handed means no revenue for the retailer and a negative customer experience, the critical factor is the fraction of size group i customers who buy an SKU from option j after visiting its webpage. We define the *configuration reaction coefficient*, and denote it with $f^i(C_j)$, to present this value.

LEMMA 1. *By imposing a series of practical assumptions outlined in Appendix EC.1, $f^i(C_j)$ is:*

$$f^i(C_j) := \begin{cases} 1, & \text{if } i \in C_j; \\ p^u, & \text{if } i, i-1 \notin C_j \text{ but } i+1 \in C_j; \\ p^l, & \text{if } i, i+1 \notin C_j \text{ but } i-1 \in C_j; \\ p^u + p^l, & \text{if } i \notin C_j \text{ but } \{i-1, i+1\} \subseteq C_j; \\ 0, & \text{o.w.;} \end{cases} \quad (2)$$

where $p^u, p^l \geq 0$ and $p^u + p^l \leq 1$.

One can interpret Lemma 1 and (2) as follows. If an option is presented in size i , customers in size group i choose that SKU, and there is no size stockout or size substitution effect. However, if the primary size is unavailable, some customers in size group i may substitute with one of the SKUs in the neighboring sizes, and some may leave without a purchase. The fraction of customers who substitute with a larger size (p^u) may differ from that of customers who substitute with a smaller size (p^l); therefore, in general, $p^u \neq p^l$. In practice, the size substitution behavior is localized and customers consider at most one of the neighboring sizes for substitution.

3.2. The Revenue and Unfairness Metric

With the choice model developed in Section 3.1, defining the retailer's revenue term is straightforward. Knowing that a fraction $f^i(C_j)$ of size group i customers buy an SKU from option j after visiting its webpage, the retailer's expected revenue from displaying assortment S to a size group i customer is:

$$\text{Rev}(S|i) = \frac{\sum_{j \in S} v_j f^i(C_j) r_j}{v_0 + \sum_{m \in S} v_m}. \quad (3)$$

Because in the baseline model the retailer cannot tailor the category webpage to the visiting customer's size, and size group i customers constitute fraction ξ^i of customers, the expected revenue from displaying assortment S is:

$$\text{Rev}(S) = \sum_{i \in G} \xi^i \frac{\sum_{j \in S} v_j f^i(C_j) r_j}{v_0 + \sum_{m \in S} v_m} = \frac{\sum_{j \in S} v_j \tilde{r}_j(C_j)}{v_0 + \sum_{m \in S} v_m}, \quad (4)$$

where $\tilde{r}_j(C_j) := r_j \sum_{i \in G} \xi^i f^i(C_j)$.

We now introduce a metric to measure unfairness in customer experience. To do so, we need an index of the experiences of different customer size groups with respect to size availability. In Section 3.1, we noted that a fraction $(1 - f^i(C_j))$ of size group i customers who visit option j 's webpage do not purchase an SKU because of size unavailability. Therefore, we define unavailability, $U^i(S)$, as the fraction of size group i customers who leave the store empty-handed:

$$U^i(S) := \frac{v_0}{v_0 + \sum_{m \in S} v_m} + \frac{\sum_{j \in S} v_j (1 - f^i(C_j))}{v_0 + \sum_{m \in S} v_m}. \quad (5)$$

From the retailer's perspective, $U^i(S)$ is a practical measure because the probability that a customer leaves the store empty-handed can be estimated from clickstream data. Moreover, customers often evaluate assortment fairness based on whether they (compared to others) can find and purchase a suitable option, which $U^i(S)$ captures. Finally, the methods we develop under $U^i(S)$ can be readily adapted to any performance metric of the form $(\sum_{j \in S} \phi_j^i v_j) / (1 + \sum_{j \in S} v_j)$, where ϕ_j^i is an arbitrary value. For example, the measure $(\sum_{j \in S} \mathbf{1}(i \in C_j) v_j) / (1 + \sum_{j \in S} v_j)$ can be used when customers are likely to purchase an option only if it is available in their primary size and do not substitute with neighboring sizes.

Unfairness in a firm's operations can be measured in various ways. One approach is to consider the gap between the payoffs received by the best-off and worst-off customers. This is the *pairwise fairness* measure applied in designing recommendation algorithms in Beutel et al. (2019) and Narasimhan et al. (2020). Most relevant to our work, Chen et al. (2022) apply this measure in an assortment optimization setting, where an online platform ensures that its different sellers receive a fair share of visibility on the retailer's website. The popularity of pairwise fairness stems from its straightforward interpretation and ease of integration into mathematical models (Karsu and Morton 2015). Following this line of logic, we use the gap between the highest and lowest discomfort among size groups as the unfairness metric for an assortment:

$$\text{Unf}(S) := \max_{l, k \in G} [U^k(S) - U^l(S)]. \quad (6)$$

An alternative measure is the *Rawlsian* metric, whereby the maximum discomfort represents the level of unfairness in a system (Farajollahzadeh and Hu 2021); we discuss this metric in Section 5.5.

3.3. The Objective Function

There are two common approaches to incorporating behavioral considerations into an optimization problem. In one approach, the anticipated behavioral consequences of a pricing or assortment decision are embedded in customer utility and the aggregate demand function (Popescu and Wu 2007, Wang 2018, Hu and Nasiry 2018). In a second approach, the firm (or a central planner) imposes constraints on pricing and assortment decisions to satisfy regulatory requirements and/or prevent (or limit) customer backlash and reputational damage from practices perceived as unfair or unequal (Sumida et al. 2021, Cohen et al. 2022, Li et al. 2024), with the goal of ensuring inclusivity. Our notion of fairness aligns with the second approach and is closely related to inclusivity in retail practices because it aims to reduce the systematic exclusion of certain size groups from being served by the assortment. At the same time, it is conceptually distinct from several related ideas. First, our metric does not model customer satisfaction or disappointment directly, which would require a utility-based choice framework and the estimation of additional behavioral parameters. Second, it differs from accessibility: in our model, all customers have access to the assortment, and the issue we address is unequal service outcomes conditional on access. Finally, our fairness concept differs from equity. While equity emphasizes differential treatment based on heterogeneity across size groups in the customer base, our baseline metric penalizes disparities symmetrically across size groups.

Unfair practices in assortment decisions have two possible effects on the retailer. First, social media makes it easier for customers to become aware of non-inclusive and unfair practices. Second, such perceived unfairness may influence customers' future purchase decisions. As Bateni et al. (2022) put it, fairness is a "crucial system objective" in settings where a profit-maximizing platform seeks to ensure repeat purchases by treating customers fairly. Therefore, given the competitive fashion retail landscape where customer retention is a strategic concern, we define the retailer's *profit function* as follows:

$$\Pi(S) := \text{Rev}(S) - \lambda \text{Unf}(S), \quad (7)$$

where $\lambda \geq 0$ represents the relative importance of unfairness concerns compared to revenue. As λ increases, the retailer places more emphasis on fairness in selecting its assortment. As discussed in the Introduction, customers form their loyalty decision during their first interactions with the website, and a negative impression created by size unavailability can therefore lead to attrition. As such, λ can be interpreted as the expected reduction in the retailer's profit due to customer attrition.¹

¹ An alternative interpretation of λ is as the Lagrange multiplier of a hard-constraint formulation in which the retailer maximizes expected revenue subject to $\max_{l,k \in G} [U^k(S) - U^l(S)] \leq F_{\max}$, where $F_{\max}$ is the largest tolerable gap in unavailability across size groups. By varying $F_{\max}$ and computing the associated multiplier λ^F , the retailer can trace a correspondence between its fairness tolerance and the implied value of λ . This correspondence offers a practical calibration strategy: to enforce a desired fairness tolerance (e.g., $F_{\max} = 0.10$), the retailer can solve the hard-constraint problem at that tolerance and use the resulting λ^F in the soft-penalty formulation (7).

We write the retailer's deterministic assortment optimization problem subject to the largest gap in the measure (5) as:

$$\begin{aligned} \Pi_D = \max_S \quad & \text{Rev}(S) - \lambda \max_{l,k \in G} [U^k(S) - U^l(S)] \\ \text{s.t.} \quad & S \subseteq [n], \end{aligned} \quad (\text{AOP-DetFair})$$

where (AOP-DetFair) stands for Assortment Optimization Problem - Deterministically Fair, and Π_D denotes the optimal profit of a deterministic approach.

Whereas (AOP-DetFair) aligns with the classic assortment optimization problems in a physical setting, in online retailing, which is our focus, it takes an additional layer. Unlike a physical retailer, an online retailer is not limited to offering only one assortment to all arriving customers, and can *randomize* over a set of assortments, meaning it can choose a probability distribution over the feasible assortments and display an assortment sampled from this distribution. With the rise of complex optimization methods in online retailing, randomization in assortment optimization has gained traction in recent years (see e.g., Chen et al. (2022), Wang et al. (2024), Zhu et al. (2025)). For instance, Wang et al. (2024) apply it to robustify the retailer's revenue against estimation errors. More recently, Zhu et al. (2025) employ randomization to solve an assortment optimization with different suppliers where market share constraints must be satisfied.

With randomization, the retailer decides $\mathbf{P}$, a vector of length 2^n representing the display probability of each feasible assortment; which means $\mathbf{P} \geq \mathbf{0}$ and $\mathbf{P}^\top \mathbf{1} = 1$. We denote the probability of displaying assortment S with $P(S)$. With randomization, we define (in italics) $\text{Rev}(\mathbf{P}) := \sum_{w=1}^{2^n} P(S^w) \text{Rev}(S^w)$ and $U^i(\mathbf{P}) := \sum_{w=1}^{2^n} P(S^w) U^i(S^w)$ to display the *expected revenue* and *expected discomfort*, respectively. Using the two, we define the retailer's randomized assortment optimization problem as:

$$\begin{aligned} \Pi_R = \max_{\mathbf{P} \geq \mathbf{0}} \quad & \text{Rev}(\mathbf{P}) - \lambda \text{Unf}(\mathbf{P}) \\ \text{s.t.} \quad & \mathbf{P}^\top \mathbf{1} = 1, \end{aligned} \quad (\text{AOP-RandFair})$$

where $\text{Unf}(\mathbf{P}) := \max_{l,k \in G} [U^k(\mathbf{P}) - U^l(\mathbf{P})]$. The name of (AOP-RandFair) stands for Assortment Optimization Problem - Randomized Fair, and Π_R denotes the optimal profit of a randomized approach. We now prove a structural property that (AOP-DetFair) and (AOP-RandFair) have in common.

LEMMA 2. *In (AOP-DetFair) and (AOP-RandFair): (i) The unfairness term is positive. (ii) Profit, revenue, and unfairness decrease in λ in the optimal solution.*

Proof: Proofs are in Appendix EC.2.

The first part of Lemma 2 is intuitive given the notion of pairwise fairness in our model. As λ becomes larger, more focus is allocated to decreasing the unfairness term, thereby decreasing both revenue and unfairness. Obviously, then, profit decreases in λ .

Randomization allows (AOP-RandFair) to attain higher profits when compared to (AOP-DetFair). Example 1 below illustrates the value of randomization in the context of our problem.

EXAMPLE 1. Suppose there are two customer groups l and k and two feasible assortments S^1 and S^2 with $\text{Rev}(S^1) = 10$ and $\text{Rev}(S^2) = 8$. Also assume that $U^l(S^1) = 0.75$, $U^k(S^1) = 0.25$, $U^l(S^2) = 0.25$, and $U^k(S^2) = 0.75$. Note that $U^k(S^1) - U^l(S^1) = -0.5$ and $U^k(S^2) - U^l(S^2) = 0.5$. Define $\pi = P(S^1)$ and observe that the expected profit is:

$$\Pi_R(\pi) = 10\pi + 8(1 - \pi) - \lambda \max\{0.5\pi - 0.5(1 - \pi), -0.5\pi + 0.5(1 - \pi)\}. \quad (8)$$

We solve (8) for two values of λ . If $\lambda = 1$, the retailer's optimal decision is $\pi^*(\lambda = 1) = 1$, which is equivalent to solely displaying S^1 . Therefore, the solution to (AOP-DetFair) and (AOP-RandFair) with $\lambda = 1$ are the same, with $\Pi_R^* = \Pi_D^* = 9.5$. Nonetheless, with $\lambda = 3$, the optimal decision is $\pi^*(\lambda = 3) = 0.5$. In this situation, we have $\Pi_R^* = 9$ while $\Pi_D^* = 7$, representing the advantage of randomization.

To see this more clearly, note that if only S^1 or only S^2 is shown, the impact of the unfairness term in (8) is equal to -0.5λ . Therefore, solely displaying S^2 is never optimal for the retailer, as assortment S^1 has the same level of unfairness as S^2 but yields more revenue. For $\lambda = 1$, setting $\pi < 1$ and displaying S^2 with a non-zero probability causes a decrease in the revenue that has a stronger effect on the profit than alleviating the unfairness. Nevertheless, for $\lambda = 3$, the impact of the unfairness term on the profit is more pronounced, and the retailer prefers to incur a drop in its revenue through randomization. ■

In Example 1, the optimal solution for (AOP-DetFair) suffers from a *systematic discrimination* toward one of the two customer groups. However, with randomization and displaying both assortments with non-zero probabilities (here, 50%,) the unfairness in one assortment is canceled out by the advantage in the other assortment. This way, the systematic discrimination pattern gives way to a random pattern. Let us define fairness as the complement of the unfairness term (i.e., $\text{Fair}(\mathbf{P}) := 1 - \text{Unf}(\mathbf{P})$). We formally discuss the impact of randomization in the following lemma.

LEMMA 3. *With randomization: (i) The unfairness term is convex in $\mathbf{P}$ while the revenue term is linear in it. (ii) The revenue-fairness frontier is concave with respect to the origin.*

The first part of Lemma 3 shows that randomizing across multiple assortments yields an expected unfairness that improves upon the worst unfairness among them, while expected profit remains their weighted average. Combined with Lemma 2, this implies a concave revenue–fairness frontier, as established in the second part of Lemma 3. We provide more insights in Section 6.2.

4. Characterization of the Optimal Solution

In this section, we characterize the structure of the optimal solution to (AOP-RandFair). We first study the special case without a broken assortment, and then develop a solution method for the general case with a broken assortment.

4.1. Optimal Solution without a Broken Assortment

Without a broken assortment, the retailer can present every SKU of every option it so desires. A classic result in the AOP literature (Talluri and Van Ryzin 2004) is that the option assortment that maximizes $(\sum_{j \in S} r_j v_j) / (v_0 + \sum_{j \in S} v_j)$ follows a nested revenue-ordered structure. Suppose this problem's optimal assortment is the set $S^{nb} = \{1, \dots, m\}$. We establish the following result on the optimal assortment in the absence of a broken assortment.

LEMMA 4. *Without a broken assortment, $P(S^{nb}) = 1$ is an optimal assortment for (AOP-RandFair).*

Because $P(S^{nb}) = 1$ is a singleton, a direct result of Lemma 4 is that displaying S^{nb} is also an optimal solution to (AOP-DetFair). Lemma 4 offers an important managerial insight: if retailers could maintain non-broken assortments, they can fully align revenue maximization with unfairness-aware profit maximization.

4.2. Optimal Solution with a Broken Assortment

Size unavailability is endemic to fashion retailing (*Retail-Week* 2022), and a non-broken assortment is rarely observed in practice. Therefore, to solve (AOP-RandFair) under a broken assortment, we introduce a framework called the *selection-based* method. This method comprises two parts: a *forward transformation*, which converts the original combinatorial assortment problem into an LP in terms of selection probabilities, and a *backward transformation*, which maps the LP solution back to an optimal randomized assortment policy.

We begin the forward transformation phase by linearizing the unfairness metric using an auxiliary variable ψ that upper-bounds all pairwise fairness differences. This yields the following equivalent formulation:

$$\begin{aligned}
 \Pi &= \max_{P, \psi} \sum_{S \subseteq N} P(S) \frac{\sum_{j \in S} v_j \tilde{r}_j(C_j)}{v_0 + \sum_{m \in S} v_m} - \psi \\
 \text{s.t. } \quad &\psi \geq \lambda \sum_{S \subseteq N} P(S) \left[\frac{v_0 + \sum_{j \in S} v_j (1 - f^k(C_j))}{v_0 + \sum_{m \in S} v_m} - \frac{v_0 + \sum_{j \in S} v_j (1 - f^l(C_j))}{v_0 + \sum_{m \in S} v_m} \right] \quad \forall k, l \in G, \\
 &\sum_{S \subseteq N} P(S) = 1, \\
 &P(S) \geq 0 \quad \forall S \subseteq N,
 \end{aligned}
 \tag{AOP-Assortment}$$

where (AOP-Assortment) stands for Assortment Optimization Problem - Assortment-Based. It is straightforward to show that without fairness considerations (i.e., $\lambda = 0$), the optimal solution is a singleton with a $\tilde{r}_j$ -ordered structure. However, in the broken-assortment case, option revenues must be adjusted to account for size availability. Therefore, the main complexity is incorporating fairness considerations.

Although this step removes the max operator from the objective, the formulation remains computationally challenging for two reasons: (i) the decision variable $P(S)$ is defined over exponentially many assortments, and (ii) the revenue term is fractional in these variables. To address both challenges simultaneously, we replace the assortment-based decision variables (i.e., the probabilities of showing individual assortments)

with option-selection probabilities. Specifically, for each option $j \in N$, we define x_j as the probability that a customer selects option j , and let x_0 denote the probability of selecting the outside option. The total-probability constraint in the new formulation is $\sum_{j \in N} x_j + x_0 = 1$. Under this reformulation, expected revenue is $\sum_{j \in N} \tilde{r}_j(C_j)x_j$, and the fairness constraints become:

$$\psi \geq \lambda \sum_{j \in N} x_j (f^k(C_j) - f^l(C_j)) \quad \forall k, l \in G.$$

Finally, to ensure that selection probabilities align with the structure imposed by an MNL choice model, we want to set x_j so that $x_j = \sum_{S \subseteq N} P(S) \Pr(j|S)$, where $\Pr(j|S)$ follows the structure in (1). For any $j \in N$ and assortment S , the probability of choosing option j is $\Pr(j|S) = v_j / (v_0 + \sum_{m \in S} v_m)$ if $j \in S$, and zero otherwise. In contrast, regardless of the assortment, the probability of choosing the outside option is $\Pr(0|S) = v_0 / (v_0 + \sum_{m \in S} v_m)$. Therefore, if the optimal solution consists of displaying a single assortment S (i.e., $P(S) = 1$), then the following relation between x_j and x_0 holds:

$$x_j = \Pr(j | S) = \begin{cases} \frac{v_j}{v_0} \Pr(0 | S) = \frac{v_j}{v_0} x_0, & \text{if } j \in S; \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

With randomization, the gap between x_j and $(v_j/v_0)x_0$ can only increase, as option j may be absent from some assortments, while the outside option is part of all assortments. As such, the correct scaling between purchase and no-purchase probabilities is enforced by

$$\frac{x_j}{v_j} \leq \frac{x_0}{v_0}, \quad \forall j \in N. \quad (10)$$

Combining these components yields the following LP:

$$\begin{aligned} \Pi^{\text{Selection-Based}} &:= \max_{\mathbf{x}, \psi} \sum_{j \in N} \tilde{r}_j(C_j)x_j - \psi \\ \text{s.t.} \quad &\psi \geq \lambda \sum_{j \in N} x_j (f^k(C_j) - f^l(C_j)) \quad \forall k, l \in G, \\ &\frac{x_j}{v_j} \leq \frac{x_0}{v_0} \quad \forall j \in N, \\ &\sum_{j \in N} x_j + x_0 = 1, \\ &x_j \geq 0 \quad \forall j \in N \cup \{0\}, \end{aligned} \quad (\text{AOP-Selection})$$

where (AOP-Selection) stands for Assortment Optimization Problem - Selection-Based.

This LP has a polynomial number of variables and constraints and can be solved efficiently using standard solvers. Methods for linearizing and solving fractional optimization problems date back to Charnes and Cooper (1962). Nevertheless, showing that the optimal selection probabilities from (AOP-Selection) can be translated into actionable assortment-based decisions is nontrivial. To establish full equivalence with the original assortment-based formulation, we use a backward transformation method developed in Topaloglu

(2013) and Gallego et al. (2015), and tailored to constrained assortment optimization problems with randomization in Zhu et al. (2025).

Let $\hat{x}_j, j \in N$, be the optimal solution to (AOP-Selection). An option can have a higher selection probability not because it must be displayed more, but because it has a higher preference weight. We account for this effect by dividing $\hat{x}_j$ by v_j . Re-index the options to have $\hat{x}_1/v_1 \geq \hat{x}_2/v_2 \geq \dots \geq \hat{x}_n/v_n$. By constraint (10), $\hat{x}_0/v_0 \geq \hat{x}_1/v_1 \geq \hat{x}_2/v_2 \geq \dots \geq \hat{x}_n/v_n$. Now define nested sets $S_m := \{1, \dots, m\}$, with $S_0 = \emptyset$. The critical step is to assign the correct display probability to each set. Let $\hat{P}$ be a candidate solution to (AOP-Assortment). Set $\hat{P}(S) = 0$ for any assortment $S \notin \{S_0, S_1, \dots, S_n\}$. We must design $\hat{P}$ so that

$$\sum_{m=0}^n \hat{P}(S_m) \Pr(j | S_m) = \hat{x}_j, \quad \forall j \in N \cup \{0\}. \quad (11)$$

Start with option n , because it appears only in S_n . By equation (11), $\hat{P}(S_n)$ is uniquely defined:

$$\hat{P}(S_n) = \frac{\hat{x}_n}{v_n} \left[v_0 + \sum_{m \in S_n} v_m \right].$$

Move to option $n-1$ which appears in two assortments S_{n-1} and S_n . By equation (11):

$$\hat{P}(S_{n-1}) \frac{v_{n-1}}{v_0 + \sum_{m \in S_{n-1}} v_m} + \hat{P}(S_n) \frac{v_{n-1}}{v_0 + \sum_{m \in S_n} v_m} = \hat{x}_{n-1}$$

or equivalently:

$$\hat{P}(S_{n-1}) = \left(\frac{\hat{x}_{n-1}}{v_{n-1}} - \frac{\hat{x}_n}{v_n} \right) \left[v_0 + \sum_{m \in S_{n-1}} v_m \right].$$

We follow the same approach for all options and obtain:

$$\hat{P}(S_j) = \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{i \in S_j} v_i \right] \quad \forall j \in N \cup \{0\}, \quad (12)$$

where we define $\hat{x}_{n+1} = 0$ for notational convenience.

By showing that the optimal solutions of (AOP-Assortment) and (AOP-Selection) are feasible for each other, and using the transformed probabilities $\hat{P}$, we prove the following theorem.

THEOREM 1. (i) *The optimal values of (AOP-Assortment) and (AOP-Selection) are equal.* (ii) *Allocation $\hat{P}$ attains the optimal solution of (AOP-Assortment).* (iii) *Expected revenue and unfairness are identical in the two formulations.*

Part (iii) of Theorem 1 sheds light on the analytical value of the selection-based methodology. If the retailer is only interested in the value of the revenue or the unfairness term (rather than its assortment-based implementation), for instance if the retailer is evaluating the effect of different values of λ , it can get the optimal revenue and unfairness values without doing the backward stage.

To gain further insights into the value of incorporating fairness considerations into the optimization model, we ran several simulations calibrated on real data, which we elaborate on in Section 6. Nonetheless, we observed in our simulations that unfairness can be drastically reduced with only a small drop in revenue. For that, we compared the outcomes of the profit-maximizing assortment and the revenue-maximizing (i.e., $\lambda = 0$) assortment across 100 randomly generated inventory samples. With $\lambda = 0.1 \max_{j \in N} r_j$, on average, the unfairness dropped by 29.10% with only 1.31% drop in the revenue; with $\lambda = 0.2 \max_{j \in N} r_j$, an average drop of 54.62% in unfairness was possible with 4.78% revenue sacrifice.

We now present structural results on the optimal assortment $\hat{\mathbf{P}}$ for (AOP-Assortment), building on the optimal solution to (AOP-Selection).

LEMMA 5. *There exists an optimal $\hat{\mathbf{P}}$ with at most g^2 distinct assortments.*

Regardless of optimality, if a solution suggests randomizing over a large number of assortments, it is impractical. Looking at (12), one may fear that $\hat{\mathbf{P}}$ may contain n distinct assortments. Lemma 5 addresses this concern by showing that $\hat{\mathbf{P}}$ needs at most g^2 distinct assortments. Although the nested construction generates up to n candidate assortments, whenever adjacent products share the same $\hat{x}/v$, the corresponding nested assortments represent the same treatment tier and thus collapse into a single effective assortment. Consequently, the support size is determined not by the number of products, but by the number of distinct breakpoints in $\{\hat{x}_0/v_0, \hat{x}_1/v_1, \hat{x}_2/v_2, \dots, \hat{x}_n/v_n\}$. Because an optimal basic feasible solution to Problem (AOP-Selection) needs at most g^2 such breakpoints to balance all pairwise fairness constraints, the induced randomized assortment policy requires at most g^2 distinct assortments.

5. Extensions

In this section, we introduce several practical extensions and show how the method developed in Section 4 can be adapted to characterize the optimal solution in each case.

5.1. Size-Filtering Possibility

Our baseline model assumes that customers first choose an option and then evaluate the sizes available for that option. However, some online retailers allow shoppers to filter by size before browsing. In this approach, customers first select a size, then choose among the available products. We call this the ‘size-filtering’ or ‘size-first’ model, interchangeably.

Before introducing the size-filtering model, it is useful to clarify how unfairness differs from the baseline. In the baseline, unfairness arises when a customer selects a product shown on the category page, only to find that it is unavailable in her size. In the size-filtering model, customers choose their size first and then browse the products available in that size. Here, unfairness arises because some size groups may face much smaller choice sets than others.²

² A customer expressed this frustration as follows: “(before filtering by size) The little button says Review 250 items, you click L & XL, button now says: Review 47 items. Like seriously?!?” (Reddit 2024).

In the size-filtering approach, the customer chooses her size (e.g., via a size-filter combobox) after entering the category webpage. The choice among the options (available in the preferred size) follows an MNL selection process. The customer then buys an SKU from the selected option with probability $f^i(C)$. By choosing her size first, and in contrast to the baseline model, the customer allows the retailer to curate the assortment based on the revealed size preference. We denote the assortment curated to customers in size group i with P^i . The retailer's problem is then to design and offer g distributions: $\mathcal{P} := \{P^1, \dots, P^g\}$. Because each size group faces an assortment curated to their size, the expected revenue from assortment distribution $\mathcal{P}$ is:

$$\text{Rev}(\mathcal{P}) = \sum_{i \in G} \xi^i E_{P^i} \text{Rev}(S | i) = \sum_{i \in G} \xi^i \sum_{S \subseteq N} P^i(S) \text{Rev}(S | i). \quad (13)$$

Similarly, the expected unfairness from assortment distribution $\mathcal{P}$ is:

$$\text{Unf}(\mathcal{P}) = \max_{l, k \in G} [E_{P^k} U^k(S) - E_{P^l} U^l(S)] = \max_{l, k \in G} \left[\sum_{S \subseteq N} P^k(S) U^k(S) - \sum_{S \subseteq N} P^l(S) U^l(S) \right]. \quad (14)$$

Putting these components together, the retailer's assortment problem under size-first customer search behavior can be written as:

$$\begin{aligned} \Pi_{\text{Size-Filtering}} &= \max_{\mathcal{P}, \psi} \sum_{i \in G} \xi^i E_{P^i} \text{Rev}(S | i) - \psi \\ \text{s.t. } \psi &\geq \lambda [E_{P^k} U^k(S) - E_{P^l} U^l(S)] \quad \forall k, l \in G, \\ \sum_{S \subseteq N} P^i(S) &= 1 \quad \forall i \in G, \\ P^i(S) &\geq 0 \quad \forall i \in G, S \subseteq N, \end{aligned} \quad (\text{SFP-Assortment})$$

where (SFP-Assortment) stands for *Size-Filtering Problem - Assortment-Based*. Because in this problem the retailer must find the optimal assortment for each customer group, the number of decision variables is $g \times 2^n$. To solve the problem efficiently, we need to apply a transformation similar to that used in the baseline model. Specifically, we introduce choice probability vectors $\vec{X} := \{X^1, \dots, X^g\}$, where $X^i := \{x_1^i, \dots, x_j^i, \dots, x_n^i\}$ represents the probability vector that a customer in size group i chooses option j in product set N . The probability of no-purchase from a customer in size group i is denoted by x_0^i . We follow the same steps as the baseline model to develop the selection-based problem:

$$\begin{aligned} \Pi_{\text{Size-Filtering}}^{\text{Selection-Based}} &:= \max_{\mathbf{X}, \psi} \sum_{i \in G} \xi^i \sum_{j \in N} r_j f^i(C_j) x_j^i - \psi \\ \text{s.t. } \psi &\geq \lambda \left[\sum_{j \in N} f^k(C_j) x_j^k - \sum_{j \in N} f^l(C_j) x_j^l \right] \quad \forall k, l \in G, \\ \frac{x_j^i}{v_j} &\leq \frac{x_0^i}{v_0} \quad \forall j \in N, \forall i \in G, \\ \sum_{j \in N} x_j^i + x_0^i &= 1 \quad \forall i \in G, \\ x_j^i &\geq 0 \quad \forall j \in N \cup \{0\}, i \in G, \end{aligned} \quad (\text{SFP-Selection})$$

where (SFP-Selection) stands for Size- Filtering Problem - Selection-Based. The above LP determines the optimal (profit-maximizing) selection probabilities for each size group. These selection probabilities can be converted into randomized assortments using a method similar to that in (12). Let $\hat{\mathcal{P}} := \{\hat{P}^1, \dots, \hat{P}^g\}$ denote the resulting assortment policy. The following theorem establishes the validity of this transformation by showing that the resulting assortment policy obtained from the LP program in (SFP-Selection) also maximizes the original assortment problem in (SFP-Assortment) and vice versa.

THEOREM 2. (i) *The optimal values of (SFP-Assortment) and (SFP-Selection) are equal.* (ii) *Allocation $\hat{\mathcal{P}}$ attains the optimal solution of (SFP-Assortment).* (iii) *Expected revenue and unfairness are identical in the two formulations.*

Note that the baseline model is a more constrained case of the size-first approach. This becomes clear by adding the set of constraints $\mathbf{P}^l = \mathbf{P}^k$ for every two size groups to (SFP-Assortment), which means the retailer displays the same assortment to all customers regardless of their selected sizes. Therefore, it is clear that the optimal profit of the size-first approach exceeds the baseline, either due to higher revenue, lower unfairness, or both. In our simulations, when comparing the results of the size-filtering model with those of the baseline model for 100 samples, we observed that allowing the customers to choose their sizes first increases the revenue by 11.80% and drops the unfairness by 51.81% on average for $\lambda = 0.1 \max_{j \in N} r_j$.

5.2. Option Substitution

In this extension, we incorporate the possibility that customers may revisit the option page after facing size unavailability for the option they first selected. That is, customers may substitute with other options in case of size stockouts. Implementing this possibility requires extending the customer choice model, the retailer's assortment decisions, and the revenue and unfairness terms. We call this model 'option-substitution' or 'revisit behavior' interchangeably.

In the baseline model, a customer from size group i selects option j and evaluates configuration C_j . If C_j contains her preferred size, she purchases an SKU and exits. Otherwise, she substitutes to neighboring sizes with probability $f^i(C_j)$ or leaves the store with probability $1 - f^i(C_j)$. In the revisit model, if the chosen option does not contain her preferred size, she returns to the category page with probability δ and repeats the same process. Hence, after the first visit, she purchases from option j with probability $f^i(C_j)$, revisits with probability $\delta(1 - f^i(C_j))$, and leaves with probability $(1 - \delta)(1 - f^i(C_j))$.

In the baseline model, the retailer offers a single assortment throughout the customer's visit. With revisits, however, the retailer can show a different assortment when the customer returns to search for another option. Because the retailer observes the customer's initial choice, this follow-up assortment can depend on that information. We therefore model a two-stage assortment policy: a first-stage randomized assortment with distribution $\mathbf{P}_1$, and a second-stage randomized assortment with distribution $\mathbf{P}_{2,(j)}$, shown to customers who return after not purchasing option j in the first stage.

Under the option-substitution model, customers may observe two assortments over their search cycle. As a result, unfairness and revenue must be calculated over the entire search cycle. We therefore define two-stage unavailability for size group i with $U^k(S_1, S_2)$, which represents the level of unavailability for a customer in size group i who is faced with assortment S_1 in the first stage and S_2 in the second stage. Similarly, we define $\text{Rev}^i(S_1, S_2)$ to denote the expected revenue generated when a customer group of size i is exposed to assortments S_1 and S_2 during her search cycle. The full expression for two-stage unavailability and revenue is available in Appendix EC.3.

Whereas the first-stage assortment distribution is the same among all customers, the second-stage distribution depends on the first option a customer visits. Therefore, the retailer must decide $\mathcal{P} := \{P_1, P_{2,(1)}, \dots, P_{2,(n)}\}$, which entails $(1 + n) \times 2^n$ decision variables. Following the pairwise approach for estimating unfairness of assortments, the retailer's assortment optimization problem with revisits can be written as:

$$\begin{aligned} \Pi_{\text{Revisit}} = & \max_{\psi, P_1, P_{2,(1)}, \dots, P_{2,(n)}} \sum_{i \in G} \xi^i E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} \text{Rev}^i(S_1, S_2) - \psi \\ \text{s.t. } & \psi \geq \lambda \left[E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} [U^k(S_1, S_2) - U^l(S_1, S_2)] \right] \quad \forall k, l \in G, \\ & \sum_{S_1 \subseteq N} P_1(S_1) = 1, \quad \sum_{S_2 \subseteq N} P_{2,(j)}(S_2) = 1 \quad \forall j \in N, \\ & P_1(S_1), P_{2,(j)}(S_2) \geq 0 \quad \forall S_1 \subseteq N, S_2 \subseteq N, j \in N, \quad P_{2,(j)}(S_2) = 0 \quad \text{if } j \in S_2. \end{aligned} \quad (\text{RBP-Assortment})$$

where (RBP-Assortment) stands for *Revisit Behavior Problem - Assortment-Based*. To solve an optimization problem of this size efficiently, we apply the same selection-based transformation used in the baseline model. Specifically, we introduce a first-stage selection vector $X = \{x_1, \dots, x_n\}$ and, for the option-substitution model, a set of second-stage selection vectors $\vec{Y} = \{Y^1, \dots, Y^g\}$, where $Y^i = \{y_1^i, \dots, y_n^i\}$ for each $i \in G$. Here, y_j^i denotes the fraction of customers in size group i who visit option j in the second stage. We use index 0 to denote the no-purchase choice in both stages.

To complete the reformulation, we also introduce a vector of auxiliary variables $\vec{Q} := \{Q_1, \dots, Q_n\}$ where $Q_j := \{q_{j,0}, q_{j,1}, \dots, q_{j,n}\}$ for each $j \in N$. The role of $\vec{Q}$ is to impose the MNL structure in customers' choice in the second stage. More specifically, $q_{j,w} / (\sum_{m \in N} q_{j,m} + q_{j,0})$ denotes the fraction of customers who choose to visit option w from all customers who chose to revisit after visiting option j 's webpage.

Therefore, we introduce the selection-based transformation of (RBP-Assortment) as follows:

$$\begin{aligned}
 \Pi_{\text{Revisit}}^{\text{Selection-Based}} &:= \max_{\psi, \mathbf{x}, \hat{Y}, \mathbf{u}, \hat{Q}} \sum_{i \in G} \xi^i \sum_{j \in N} (x_j + y_j^i) f^i(C_j) r_j - \psi \\
 \text{s.t. } &\psi \geq \lambda[u^l - u^k], \quad \forall k, l \in G, \\
 &u^i = (x_0 + y_0^i) + \sum_{j \in N} (x_j + y_j^i)(1 - f^i(C_j)), \quad \forall i \in G, \\
 &\frac{x_j}{v_j} \leq \frac{x_0}{v_0} \quad \forall j \in N, \quad \frac{q_{j,m}}{v_m} \leq \frac{q_{j,0}}{v_0} \quad j \in N, m \in N, \\
 &\sum_{j \in N} x_j + x_0 = 1, \quad \sum_{m \in N} q_{j,m} + q_{j,0} = \delta x_j \quad \forall j \in N, \\
 &y_m^i = \sum_{j \in N} q_{j,m} (1 - f^i(C_j)) \quad i \in G, m \in N \cup \{0\}, \\
 &q_{m,j} \geq 0, \quad \forall m \in N, j \in N \cup \{0\}, \\
 &q_{j,j} = 0 \quad \forall j \in N, \quad x_j \geq 0 \quad \forall j \in N \cup \{0\},
 \end{aligned} \tag{RBP-Selection}$$

where (RBP-Selection) stands for Revisit Behavior Problem - Selection-Based. After solving this problem and obtaining $\hat{X}$, $\hat{Q}$, and $\hat{Y}$, we run a backward transformation similar to the one in the baseline model to obtain $\hat{P}$. The details of the backward transformation stage are available in Appendix EC.3. We then prove the following result:

THEOREM 3. (i): The optimal value of (RBP-Assortment) and (RBP-Selection) are equal. (ii) Allocation $\hat{P}$ achieves the optimal solution of (RBP-Assortment). (iii) Expected revenue and unfairness are identical in the two formulations.

To show the baseline model is a more constrained case of (RBP-Assortment), it suffices to add the constraint $P_{2,(j)}(S_2 = \emptyset) = 1$ for every option $j \in N$. This represents a case where the retailer shows an empty assortment to customers who decided to revisit. Therefore, as in the size-filtering case, it is easy to see that the optimal profit with revisits is higher than the baseline due to either higher revenue, lower unfairness, or both. When comparing the results of the option-substitution model with those of the baseline model in our simulations for 100 samples, we observed that with $\delta = 0.5$ (i.e., half of the customers willing to revisit), the revenue increases by 6.54% and the unfairness drops by 12.72% on average for $\lambda = 0.1 \max_{j \in N} r_j$. For $\lambda = 0.25 \max_{j \in N} r_j$, the revenue increases by 8.90% but the overall unfairness rises by 14.48%, indicating that the impact of revisits on unfairness is not monotonic in λ . We further investigate the rationale behind this nuanced relationship in Section 6.

5.3. Option Substitution under Size Filtering

In this extension, we combine size filtering with option substitution to form a *hybrid* model. Unlike in the option-first revisit model, the retailer observes the customer's size upon entering the category page, allowing it to tailor both the first- and second-stage assortments accordingly.

Let P_1^i denote the distribution over first-stage assortments offered to customers in size group i , and let $P_{2,(j)}^i$ denote the distribution over second-stage assortments faced by a size- i customer who selected option j in the first stage. The retailer's optimization problem under size filtering with option substitution can be written as:

$$\begin{aligned} \Pi_{\text{Hybrid}} = & \max_{\psi, P_1^i, P_{2,(1)}^i, \dots, P_{2,(n)}^i} \sum_{i \in G} \xi^i E_{P_1^i} \sum_{j \in N} E_{P_{2,(j)}^i} \text{Rev}^i(S_1, S_2) - \psi \\ \text{s.t. } & \psi \geq \lambda \left[E_{P_1^k} \sum_{j \in N} E_{P_{2,(j)}^k} U^k(S_1, S_2) - E_{P_1^l} \sum_{j \in N} E_{P_{2,(j)}^l} U^l(S_1, S_2) \right] \quad \forall k, l \in G, \quad (\text{HP-Assortment}) \\ & \sum_{S_1 \subseteq N} P_1^i(S_1) = 1 \quad \forall i \in G, \quad \sum_{S_2 \subseteq N} P_{2,(j)}^i(S_2) = 1 \quad \forall j \in N, \forall i \in G, \\ & P_1^i(S_1), P_{2,(j)}^i(S_2) \geq 0 \quad \forall S_1 \subseteq N, S_2 \subseteq N, j \in N, i \in G, \end{aligned}$$

where (HP-Assortment) stands for *Hybrid Problem - Assortment-Based*.

We apply the selection-based transformation to find the optimal assortment. Because the transformation steps and the proof of equivalence follow a similar structure to the size-filtering and option-substitution models, we relegate the details to Appendix EC.4.

Similar to the size-filtering and option-substitution models, it is easy to show that (SFP-Assortment) and (RBP-Assortment) are more constrained cases of (HP-Assortment). As such, it is unsurprising that the hybrid model's profit exceeds that of the other two models. In our simulations on 100 randomly generated inventory samples, we observed an average increase of 12.02% in revenue and a 61.03% decrease in unfairness compared to the baseline model with $\lambda = 0.1 \max_{j \in N} r_j$.

5.4. Endogenous Configuration Decisions

In the baseline model, we assume that the configuration of each option is exogenous: if an option j is included in an assortment, then all sizes contained in its configuration set C_j are displayed on the option's webpage. We now relax this assumption and extend the model to allow for *endogenous configuration decisions*, where the retailer can choose which subset of sizes to display for each option. To illustrate, suppose option j is available in sizes $\{XS, S, M, XL\}$. Under the baseline model, displaying option j implies displaying all four sizes. Under the endogenized configuration model, the retailer may instead display this option in the configuration $\{XS, XL\}$.

Allowing configuration choices increases modeling complexity substantially because the availability of each SKU affects substitution patterns across neighboring sizes. To manage this complexity, we distinguish between the *full configuration* C_j of option j and its *displayed configuration* $H_j \subseteq C_j$. In addition to deciding which options to include, the retailer now chooses how each option is displayed. We define an *extended assortment* as a tuple $(S, \mathbf{H})$, where $S \subseteq N$ denotes the set of displayed options and $\mathbf{H} := \{H_1, \dots, H_n\}$

specifies the displayed configuration of each option. We also introduce $\mathbf{C} := \{C_1, \dots, C_n\}$ as the set of full configurations of all options. The retailer's problem becomes:

$$\begin{aligned} \Pi_{\text{Endogenized}} &= \max_{\mathbf{P}, \psi} \sum_{S, \mathbf{H}} P(S, \mathbf{H}) \text{Rev}(S, \mathbf{H}) - \psi \\ \text{s.t. } \quad &\psi \geq \lambda \sum_{S \subseteq N} P(S, \mathbf{H}) [\mathbf{U}^k(S, \mathbf{H}) - \mathbf{U}^l(S, \mathbf{H})] \quad \forall k, l \in G, \\ &\sum_{S, \mathbf{H}} P(S, \mathbf{H}) = 1, \\ &P(S, \mathbf{H}) \geq 0 \quad \forall S \subseteq N, \mathbf{H} \subseteq \mathbf{C}, \quad P(S, \mathbf{H}) = 0 \quad \text{if } \mathbf{H} \notin \mathbf{C}, \end{aligned} \tag{ECP-Assortment}$$

where (ECP-Assortment) stands for Endogenized Configuration Problem - Assortment-Based. In this problem, the constraint $\mathbf{H} \notin \mathbf{C}$ ensures feasibility by requiring that each displayed configuration satisfies $H_j \subseteq C_j$. We extend the selection-based LP approach to accommodate endogenous configurations. Since the configuration of an option does not affect its selection probability in the baseline model, we can decouple the probability that customers select an option from the probability that it is seen in a given configuration. We call this the ‘selection-configuration decoupling property’. To see that, let x_j denote the fraction of customers who select option j , and $z_{j,H}$ denote the fraction of customers who select option j and observe it in configuration H .³ The resulting linear program is:

$$\begin{aligned} \Pi_{\text{Selection-Based}}^{\text{Endogenized}} &= \max_{\mathbf{x}, \mathbf{Z}, \psi} \sum_{j \in N} \sum_{H \subseteq C_j} \tilde{r}_j(H) z_{j,H} - \psi \\ \text{s.t. } \quad &\psi \geq \lambda \sum_{j \in N} \sum_{H \subseteq C_j} z_{j,H} (f^k(H) - f^l(H)) \quad \forall k \in G, \forall l \in G, \\ &\frac{x_j}{v_j} \leq \frac{x_0}{v_0} \quad \forall j \in N, \\ &\sum_{j \in N} x_j + x_0 = 1, \quad \sum_{H \subseteq C_j} z_{j,H} = x_j \quad \forall j \in N, \\ &x_j, z_{j,H} \geq 0 \quad \forall j \in N, H \subseteq C_j, \quad z_{j,H} = 0 \quad \forall j \in N, H \notin C_j, \end{aligned} \tag{ECP-Selection}$$

where (ECP-Selection) stands for Endogenized Configuration Problem - Selection-Based.

After obtaining the optimal transformed variables $\hat{\mathbf{x}}$ and $\hat{\mathbf{Z}}$, we run the backward transformation in two steps. First, by applying the transformation in (12), we obtain the configuration-agnostic display probability $\hat{P}_{CA}$, which is independent of each option's configuration decision. Then we run an iterative algorithm that uses the decoupling property to assign the optimal display probability to each configuration. The algorithm is available in Appendix EC.5. After running this algorithm, we obtain the optimal display probability $\hat{P}$. Finally, we prove the following theorem.

THEOREM 4. (i) The optimal value of (ECP-Assortment) and (ECP-Selection) are equal. (ii) Allocation $\hat{P}$ achieves the optimal solution of (ECP-Assortment). (iii) Expected revenue and unfairness are identical in the two formulations.

³ We abuse the notation by using H (which is a vector) as the index for $z_{j,H}$ to keep the notation comprehensible.

The baseline problem is a more constrained version of (ECP-Assortment). Nonetheless, since endogenizing configuration decisions can only lead to hiding an SKU from the display (despite its existence in the inventory), it cannot increase the revenue—unlike the size-filtering or option-substitution extensions. However, especially for larger values of λ , it can drastically reduce unfairness. In our simulations on 100 randomly generated inventory samples, we observed an average decrease of 1.58% in the revenue, with an average drop of 29.68% in the unfairness when compared to the baseline model, with $\lambda = 0.2 \max_{j \in N} r_j$.

5.5. Rawlsian Fairness

Rawlsian fairness focuses on the outcome of the worst-off member of a society. As such, we introduce the Rawlsian unfairness metric as the maximum unavailability among all size groups: $Unf_{\text{Rawlsian}}(\mathbf{P}) := \max_{i \in G} U^i(\mathbf{P})$. While we do not provide the technical proof here, extending the baseline model to implement the Rawlsian unfairness metric and then finding the optimal assortment distribution $\mathbf{P}_{\text{Rawlsian}}$ is straightforward. To do so, it suffices to introduce a ‘fictitious’ customer group without any impact on the expected revenue (i.e., $\xi^f = 0$) whose unavailability metric is zero for any assortment ($U^f(S) = 0$ for any assortment $S \subseteq N$).

6. Numerical Experiments

In this section, we compare profitable yet size-inclusive assortments with classic assortment optimization solutions to draw managerial insights. We first use stylized models and simulations calibrated to real data to examine the role of key operational factors. We then study how the extensions affect the results.

6.1. Simulation Setup

Using the data provided by our industry partner, we run a set of regressions to estimate model parameters. More specifically, we adopt the difference-in-differences method developed in Li et al. (2023) to estimate the substitution parameters and the distributions of v_j and r_j . In this approach, the treatment groups consist of SKUs whose neighbor sizes are out of stock, while the control group consists of SKUs whose neighbor sizes are in stock. Subsequently, we expect to see the impact of size substitution on the treatment group and not on the control group. Details of the calibration process and adaptation of the difference-in-difference method are relocated to Appendix EC.6.

Based on the dataset findings, the substitution coefficients, p^u and p^l , are set to 0.25 in the simulations. We also set $G = \{\text{XS}, \text{S}, \text{M}, \text{L}, \text{XL}, \text{XXL}\}$ and $n = 100$. The distributions of v_j, r_j, ξ^i , and the level of stockouts for each size also mimic the distributions in the dataset.

6.2. The Revenue–Fairness Frontier

As we proved in Lemma 3, the revenue–fairness frontier is concave. We plotted revenue versus unfairness, both normalized to their respective reference (maximum) values, for a sample assortment for different values

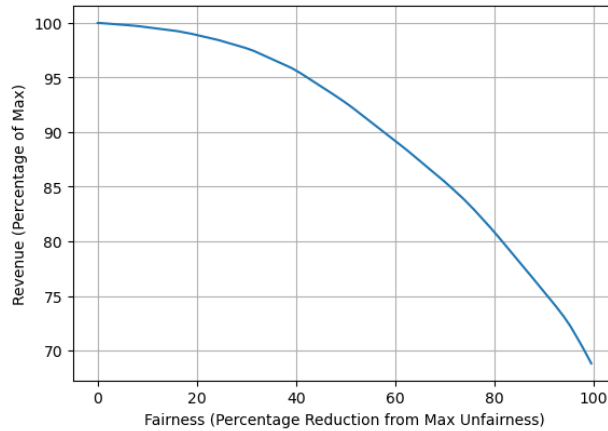


Figure 2 Revenue–fairness frontier (fairness as percentage drop in unfairness).

of λ in Figure 2. This yields a two-dimensional trade-off curve analogous to an efficient frontier in portfolio optimization (risk–return trade-off).

Moving along the curve illustrates how incremental fairness improvements (reductions in unfairness) require sacrificing some revenue. However, the relation between fairness and revenue is not linear. Initial fairness improvements can be achieved with relatively small revenue losses. However, near-perfect fairness (i.e., every size being represented exactly equally) requires increasingly larger revenue sacrifices.

6.3. The Impact of Inclusivity Considerations on the Optimal Assortment

To better illustrate how fairness affects optimal decisions, we decompose the fairness term and report unavailability rates separately for each size group. Our fairness metric controls the gap between the maximum and minimum unavailability levels across sizes.

When $\lambda = 0$, the retailer maximizes revenue only. In this case, revenue is at its highest level, but the disparity across sizes is substantial. For example, the XXL size group experiences an unavailability rate of approximately 65%, whereas the M size group experiences roughly 35%. As λ increases, the unavailability rates across sizes converge (see Figure 3), and the gap between the maximum and minimum unavailability rates shrinks. This convergence highlights a key structural insight: *fairness-oriented assortments do not eliminate unavailability, but rather redistribute it more evenly across sizes*. In other words, fairness is achieved by reducing systematic differences rather than by displaying every available size–option combination in the assortment.

To provide a concrete characterization of fairness-oriented assortments and to see the impact of such considerations on the width of an assortment, we developed a stylized example.

EXAMPLE 2. In Figure 4, we display an inventory of eight options. Each column corresponds to an option, and each row corresponds to a size. An entry of 1 indicates that a size–option pair is offered; 0 indicates that the size is unavailable for that option. For instance, Option 1 is available in all sizes, while

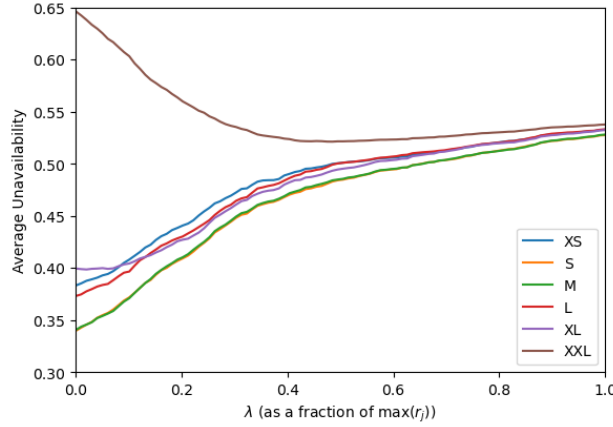


Figure 3 Unavailability rates by size group as a function of λ , averaged over 100 samples.

Option 4 is only available in sizes M and L. In this example, to isolate the impact of size coverage, we set the revenues (r_j 's) and preference weights (v_j 's) of all options equal. However, we include all possible configurations: from available in all sizes (Option 1) to available only in major sizes (Option 3) to available only in marginal sizes (Option 6), etc. Also, we chose a symmetric triangular distribution so that sizes M and L have the largest populations, and sizes XS and XXL have the smallest.

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈
XS	1	0	0	0	1	1	1	0
S	1	1	0	0	1	1	1	0
M	1	1	1	1	1	0	0	0
L	1	1	1	1	1	0	0	0
XL	1	1	0	1	0	1	0	1
XXL	1	0	0	1	0	1	0	1

Figure 4 Inventory availability of options and sizes.

We then report snapshots of optimal assortments at three representative values of λ : low ($\lambda = 0$), medium ($\lambda = 0.25$), and high ($\lambda = 0.5$) in Figure 5. The selected options are distinguished with the color green. To prevent the complexities of displaying randomization, we designed this example so that there is no need for randomization in the optimal solution.

At low λ , the '0' values are disproportionately concentrated in plus sizes (e.g., XXL), implying systematic exclusion. As λ increases, '0' values are distributed more evenly across sizes. To quantify this pattern, we compute the proportion of '0' values (i.e., the proportion of options unavailable in a customer size) for each size and report them alongside the assortment matrices. So, when $\lambda = 0$, the maximum gap between unavailability proportions is 50%, and this gap occurs between one of the marginal sizes (XXL or XS) and one of the middle sizes (M or L). However, as λ increases, the maximum gap between unavailability proportions

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈	O's
XS	1	0	0	0	1	1	1	0	50%
S	1	1	0	0	1	1	1	0	25%
M	1	1	1	1	1	0	0	0	0%
L	1	1	1	1	1	0	0	0	0%
XL	1	1	0	1	0	1	0	1	25%
XXL	1	0	0	1	0	1	0	1	50%

(a)

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈	O's
XS	1	0	0	0	1	1	1	0	40%
S	1	1	0	0	1	1	1	0	20%
M	1	1	1	1	1	0	0	0	20%
L	1	1	1	1	1	0	0	0	20%
XL	1	1	0	1	0	1	0	1	20%
XXL	1	0	0	1	0	1	0	1	40%

(b)

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈	O's
XS	1	0	0	0	1	1	1	0	25%
S	1	1	0	0	1	1	1	0	25%
M	1	1	1	1	1	0	0	0	25%
L	1	1	1	1	1	0	0	0	25%
XL	1	1	0	1	0	1	0	1	25%
XXL	1	0	0	1	0	1	0	1	25%

(c)

Figure 5 Optimal assortments for $\lambda = 0$ (a), $\lambda = 0.25$ (b), and $\lambda = 0.5$ (c).

decreases, first to 20% when $\lambda = 0.25$, and then to 0 when $\lambda = 0.50$. As this example illustrates, fairness improves not because every size is offered for every option, but because the retailer strategically reallocates size coverage by choosing which options to include in the assortment. For instance, while Option 4 does not contain S and XS sizes, Option 5 omits XL and XXL sizes, thereby balancing representation of sizes across the final assortment.

Additionally, notice that the width of the assortment (i.e., the number of items in the assortment) does not necessarily increase or decrease as λ increases. For instance, as λ increases from 0 to 0.25, the retailer increases size inclusivity by expanding the assortment and including Option 6, which is unavailable in the middle sizes. In contrast, as λ increases from 0.25 to 0.5, the retailer increases size inclusivity by shrinking the assortment and excluding Option 2, which is unavailable in the marginal sizes. ■

6.4. Impact of Size-Substitution Behavior on the Revenue-Fairness Trade-off

In Section 3.1, to model customers' size substitution behavior, we defined a customer's size substitution tolerance as t , where a customer with physical size b only considers SKUs within the size range $[b - t, b + t]$, and chooses the closest available SKU in that range. To see how different levels of size substitution impact our results, we consider three tolerance values: $t \in \{\frac{1}{2}, \frac{3}{4}, \frac{7}{4}\}$. For ease of explanation, we focus on customers in size group L who visit option j . When $t = \frac{1}{2}$, customers purchase only if $L \in C_j$ and leave otherwise, corresponding to a substitution window without any other sizes in it. When $t = \frac{3}{4}$, if $L, M \notin C_j$ but $XL \in C_j$, a quarter of customers will buy size XL, corresponding to a substitution window with at most one other size (other than the focal size) in it. When $t = \frac{7}{4}$, all customers in size group L buy an item as long as one of the sizes M, L, or XL is available. Also, if none of the sizes S, M, L, or XL are available, but size XXL is available, a quarter of them will buy size XXL, corresponding to a substitution window with at least two and at most three other sizes. We elaborate more on this example in Appendix EC.1.

Holding all other parameters fixed, increasing the substitution window mechanically increases $f^i(C_j)$ for every size group i and option j . Therefore, we expected wider substitution windows to lead to higher revenue and lower unfairness. To assess this, for a sample assortment, we redesigned all numerical figures to explicitly plot revenue versus unfairness, both normalized as percentages of their respective reference (maximum) values, for different values of λ . (Note that the reference values are obtained from finding the

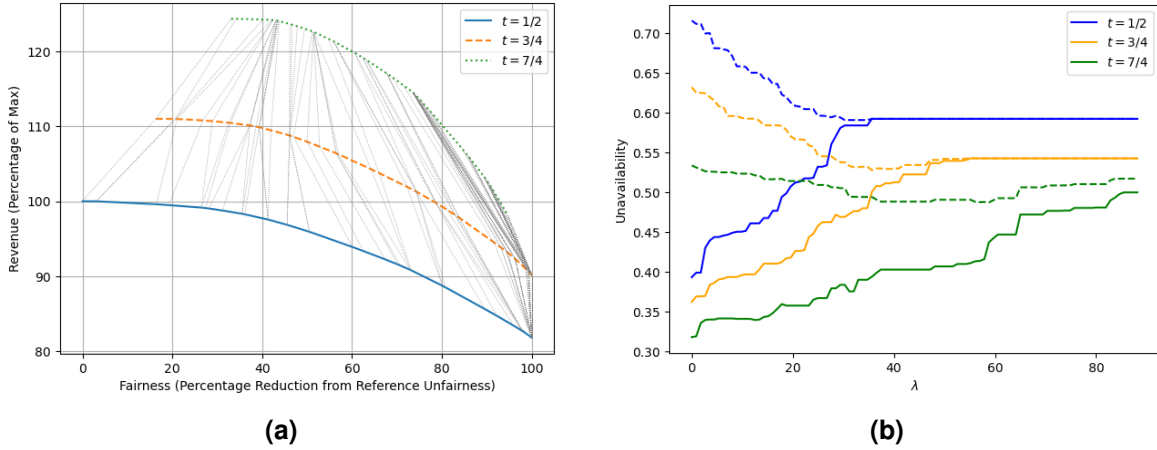


Figure 6 Panel (a): Revenue-fairness trade-off for λ ranging from 0 to $\max\{r_j\}$ for different size substitution tolerance t . Gray lines connect points with the same λ . Panel (b): The unavailability of size M (solid lines) and size XXL (dashed lines) for λ ranging from 0 to $\max\{r_j\}$ for different t .

revenue-maximizing (i.e., $\lambda = 0$) assortment in the baseline model.) This yields a two-dimensional trade-off curve analogous to an efficient frontier in portfolio optimization (risk–return trade-off).

While we indeed observe higher revenue as the substitution window widens, Figure 6a reveals a nuanced pattern in unfairness. Particularly for large values of the fairness parameter λ —corresponding to the points in the lower-right of the plot—a wider substitution window does not necessarily translate into lower unfairness. To ensure that this observation is not driven by a particular instance, we repeated the analysis across a large set of randomly generated assortments and found the same qualitative pattern.

To better understand this result, we plot the unavailability levels of size M (the best-off size group) and size XXL (the worst-off size group) as functions of λ for different substitution windows in Figure 6b. We observe that increasing the substitution window reduces unavailability for both size groups. However, the gap between the two groups closes faster when the substitution window is narrower.

We conjecture that this pattern is driven by the revenue implications of wider substitution windows. A larger window substantially increases expected revenue, which in turn raises the opportunity cost of further reducing unfairness. As a result, for large λ 's, the retailer prefers to tolerate a small but non-zero level of unfairness rather than fully equalizing discomfort across size groups. This mechanism explains why wider substitution windows do not necessarily lead to lower unfairness, despite improving overall availability.

6.5. Impact of Option-Substitution on the Retailer's Fairness Considerations

In Section 5.2, we argued that the baseline model is a more constrained version of the option-substitution model. Keeping other variables unchanged, option substitution leads to higher expected revenue by enabling additional substitution opportunities. However, its impact on unfairness is more nuanced. To investigate this, for 50 randomly generated samples, we found the optimal solution for $\lambda = 0.25 \max\{r_j\}$ under the baseline and option-substitution models.

We are interested in seeing how accounting for the possibility of revisits affects the retailer's assortment decisions. To do so, we evaluate the revenue and unfairness of the first-stage assortments generated by the option substitution model, as well as the total revenue and unfairness (first- and second-stage assortments combined). We normalize each one to the relative value in the optimal baseline assortment. The results are depicted in Figure 7.

This figure shows that although total revenue increases under the revisit model, overall unfairness does not move monotonically. More importantly, the first-stage assortments selected under the revisit model are typically more unfair than those in the baseline model. Intuitively, when the retailer anticipates that some customers will remain in the system after experiencing size unavailability, it has incentives to display more unfair assortments (that prioritize revenue) in the first stage and rely on the second stage to partially mitigate unavailability. This effect becomes stronger as the fraction δ of customers willing to revisit increases.

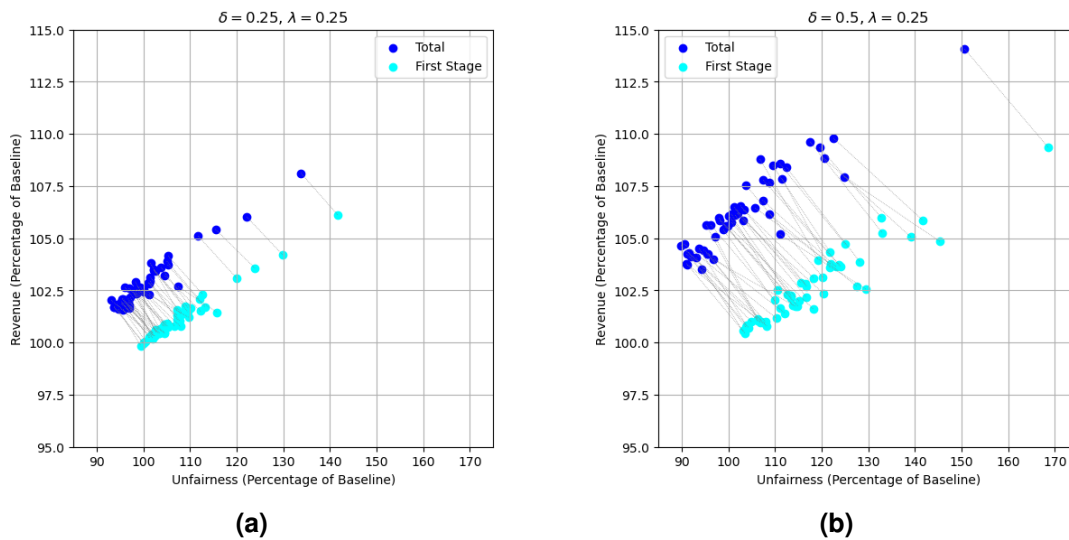


Figure 7 Comparison of the first-stage and second-stage revenue and unfairness in the option substitution approach with $\delta = 0.25$ in Panel (a) and $\delta = 0.5$ in Panel (b).

7. Discussion and Conclusion

In this paper, we modeled the static assortment problem faced by an online fashion retailer, which can display a set of assortments at random, to find a solution that balances revenue and fairness. To achieve this, we developed a choice model that incorporates the size-substitution effect and defined a pairwise fairness metric. We then employed and modified the selection-based method, which, in its forward stage, transforms the problem into a linear one that can be solved quickly, and, in the backward stage, translates the results into assortment display probabilities.

From a theoretical perspective, the novelty of our work lies in applying the selection-based method to a problem with group-wise fairness constraints, made possible by randomization. Also, we introduced a

set of practical extensions, such as embedding size filtering and option-substitution behaviors in the choice model. Moreover, extending the selection-based method, we were able to find the optimal solution for each problem. From an estimation perspective, we use a dataset provided by a partner fashion retailer and apply the DiD framework developed by Li et al. (2023) to estimate the model coefficients.

Using parameters calibrated to real data, we conducted simulations to evaluate the managerial value of our methods, yielding three key results. First, a retailer can often reduce unfairness in its product offering by nearly 30% with only a 1.5% drop in revenue. This outcome is promising, both for customers in marginalized groups and for fashion retailers seeking low-cost solutions to their size non-inclusivity problem. Second, our method tends to reduce unfairness not necessarily by reducing unavailability for all groups (which can be very costly), but by redistributing it evenly across different sizes. Finally, allowing the customer to select her size first, or encouraging her to revisit the website in case of size unavailability, increases profit by reducing unfairness, increasing revenue, or both.

Multiple extensions are possible. First, we assumed a sequential structure whereby a customer's first-stage consideration set contains only one option she chooses from the option assortment. However, customers may first build a consideration set with multiple options and sizes, then decide which option to purchase, considering each option's size availability. Second, extending the model to upstream stages (e.g., design or dynamic inventory) is promising. As argued in Section 1, size inclusivity is deeply rooted, and eradicating it requires coordinated, multi-level solutions. Finally, despite the convenience of using the MNL choice model, other choice models may better capture customer behavior in different contexts.

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Electronic Companion to “Designing Size Inclusive Fashion Assortments”

This online companion contains the appendices to the main paper.

EC.1. Choice Model: The Second Stage

We assume a customer’s physical size b follows the distribution $\eta(b)$. We define the ‘fit quality’ as a function $\text{fit}(b, i)$ to show the fit quality that a size- b customer assigns to a size- i SKU, whereby customers only consider sizes with positive fit quality. For instance, consider the function $\text{fit}(b, i) = 1 - |b - i|/t$ that constitutes a symmetric fit quality function for a size- b customer for a size i SKU with distance threshold t . With this function, the customer only considers SKU sizes with $|b - i| \leq t$.

In the second stage, upon observing C_j , the size- b customer defines her size consideration set $Z(b, C_j)$ as the set of sizes that are (i) available in C_j , and (ii) provide a positive fit quality. We assume each customer chooses the SKU size in $Z(b, C_j)$ with the highest fit quality (and nothing if $Z(b, C_j)$ is empty), and define the function $\pi(b, C_j)$ to denote the SKU size with the highest fit quality in $Z(b, C_j)$. The size of $\pi(b, C_j)$, then, is:

$$|\pi(b, C_j)| = \begin{cases} 1, & \text{if } Z(b, C_j) \text{ is not empty;} \\ 0, & \text{if } Z(b, C_j) \text{ is empty.} \end{cases}$$

As a customer purchases the SKU $\pi(b, C_j)$, the probability that a size- b customer chooses option j and then buys an SKU from it is:

$$P(j|S, b) = \frac{v_j |\pi(b, C_j)|}{1 + \sum_{k \in S} v_k}. \quad (\text{EC.1})$$

So, if a retailer offers assortment S , its expected revenue from a size- b customer is:

$$\text{Rev}(S|b) = \frac{\sum_{j \in S} v_j |\pi(b, C_j)| r_j}{1 + \sum_{k \in S} v_k}. \quad (\text{EC.2})$$

The retailer’s expected revenue from assortment S then is:

$$\text{Rev}(S) = \int_0^\infty \text{Rev}(S|b) \eta(b) db = \int_0^\infty \frac{\sum_{j \in S} v_j |\pi(b, C_j)| r_j}{1 + \sum_{k \in S} v_k} \eta(b) db. \quad (\text{EC.3})$$

We make the following assumptions to develop a closed-form formulation for (EC.3). First, SKU sizes are discrete numbers between 1 and g , that is, $G = \{1, 2, \dots, g\}$. Second, the function strictly decreases as the distance between the customer size b and the SKU size i increases. Third, the fit function is homogeneous to a shift, meaning $\text{fit}(b + x, i + x) = \text{fit}(b, i)$ for any $x \geq 0$. Fourth, for now, we assume $\text{fit}(b, i) < 0$ for $|i - b| > 1$, which implies that there are at most two sizes that provide an acceptable fit for a customer.

By the second assumption, there exists a unique $\sigma \in [0, 1]$ such that $\text{fit}(i + \sigma, i) = \text{fit}(i + \sigma, i + 1)$ for any $i \in G$, meaning a customer with size $i + \sigma$ assigns the same fit quality to SKU sizes i and $i + 1$. We can then think of ‘size group i ’ as customers whose size is in the interval $m^i := [i - 1 + \sigma, i + \sigma)$. For these customers, the primary size is i . If $i \in C_j$, then $\pi(b, C_j) = i$ for these customers because by the second assumption

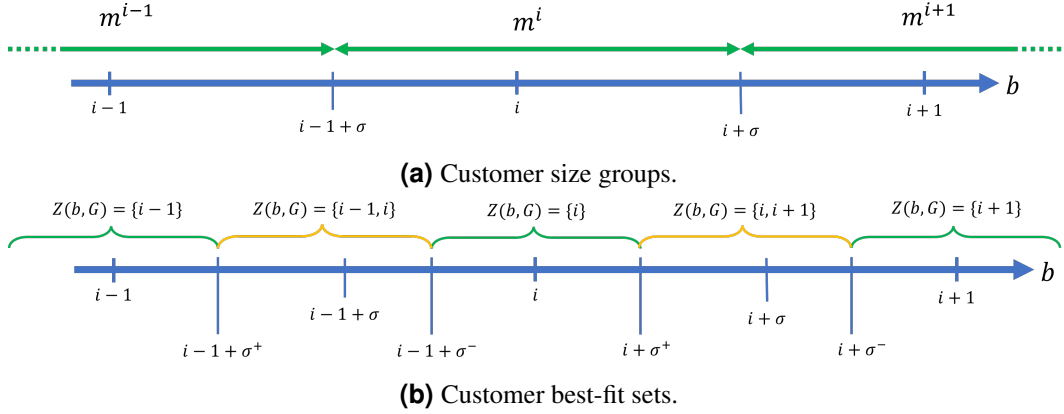


Figure EC.1 Illustration of customer size groups (m^i) and best-fit sets ($Z(b, G)$). Here, b is the continuum of customer size.

$\text{fit}(b, i) \geq \text{fit}(b, i-1)$ and $\text{fit}(b, i) \geq \text{fit}(b, i+1)$. These customers constitute a fraction $\xi^i := \int_{i-1+\sigma}^{i+\sigma} \eta(b) db$ of all customers. We define $\Xi := \{\xi^1, \dots, \xi^g\}$ as the fraction of customers in each size group. Figure EC.1a illustrates an example of customer size groups.

Beyond σ , there are two additional key values. There exists a unique $\sigma^+ \in [0, \sigma]$ such that $Z(b, G) = \{i, i+1\}$ for $b \in [i+\sigma^+, i+\sigma)$ and $Z(b, G) = \{i\}$ for $b \in [i, i+\sigma^+]$. In other words, for size group i customers whose size lies within $[i+\sigma^+, i+\sigma)$, size $i+1$ is included as the second option in $Z(b, G)$. Similarly, a unique value $\sigma^- \in [\sigma, 1]$ exists where $Z(b, G) = \{i, i-1\}$ for $b \in [i-1+\sigma, i-1+\sigma^-]$ and $Z(b, G) = \{i\}$ for $b \in [i-1+\sigma^-, i)$. Figure EC.1b provides an example illustrating the proper-fit sets for customers.

We assume $\eta(b)$ is constant within every size group. We can then replace $\eta(b)$ for $b \in m^i$ with ξ^i . Although this assumption is not necessary to establish our theoretical results, we explain later how it significantly reduces the amount of data required to estimate the model's coefficients.

We can now rewrite (EC.3) as:

$$\text{Rev}(S) = \sum_{i=1}^g \int_{i-1+\sigma}^{i+\sigma} \text{Rev}(S | b) \eta(b) db = \sum_{i=1}^g \int_{i-1+\sigma}^{i+\sigma} \sum_{j \in S} \frac{v_j |\pi(b, C_j)| r_j}{1 + \sum_{k \in S} v_k} \eta(b) db.$$

The only term in $\text{Rev}(S)$ that depends on b is $\pi(b, C_j)$. Because $\eta(b)$ is constant for $b \in m^i$, we replace it with the constant ξ^i , and write $\sum_{i=1}^g \int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| \eta(b) db = \sum_{i=1}^g \xi^i \int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| db$. Now, depending on C_j , there are five possible cases:

$$\int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| db = \begin{cases} \int_{i-1+\sigma}^{i+\sigma} 1 db = 1, & \text{if } i \in C_j; \\ \int_{i+\sigma^+}^{i+\sigma} 1 db = \sigma - \sigma^+, & \text{if } i, i-1 \notin C_j \text{ but } i+1 \in C_j; \\ \int_{i-1+\sigma}^{i-1+\sigma^-} 1 db = \sigma^- - \sigma, & \text{if } i, i+1 \notin C_j \text{ but } i-1 \in C_j; \\ \int_{i+\sigma^+}^{i+\sigma} 1 db + \int_{i-1+\sigma}^{i-1+\sigma^-} 1 db = \sigma^- - \sigma^+, & \text{if } i \notin C_j \text{ but } \{i-1, i+1\} \subseteq C_j; \\ 0, & \text{o.w.} \end{cases} \quad (\text{EC.4})$$

For ease of exposition, we define $p^u := \sigma - \sigma^+$ and $p^l := \sigma^- - \sigma$, which implies $p^u + p^l = \sigma^- - \sigma^+$. As such, we define the fraction of customers in size group i who select an SKU from option j when facing configuration C_j as *configuration reaction coefficient*, and denote it with $f^i(C_j)$ where:

$$f^i(C_j) := \int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| db = \begin{cases} 1, & \text{if } i \in C_j; \\ p^u, & \text{if } i, i-1 \notin C_j \text{ but } i+1 \in C_j; \\ p^l, & \text{if } i, i+1 \notin C_j \text{ but } i-1 \in C_j; \\ p^u + p^l, & \text{if } i \notin C_j \text{ but } \{i-1, i+1\} \subseteq C_j; \\ 0, & \text{o.w.} \end{cases} \quad (\text{EC.5})$$

To end this section, we explain why $\eta(b)$ being constant within every size group is helpful. Without this assumption, configuration reaction coefficients differ across size groups. This increases the number of coefficients from two (p^u and p^l) to $2g$ (two coefficients for every size group), significantly increasing the required data for estimations.

Explanations on Section 6.4

We use the fit quality function we introduced at the beginning of this section: $\text{fit}(b, i) := 1 - |b - i|/t$. Because it is symmetric, we have $\sigma = 1/2$. Also, due to its symmetry, we only explain substitution behavior for larger sizes; the same applies to smaller sizes.

For $t = 1/2$, $\sigma^+ = \sigma$, meaning every customer size- b has consideration set $Z(b, G) = \{i\}$. With $t = 3/4$, we have $\sigma^+ = 1/4$, meaning customers with size $b \in [i - 1/4, i + 1/4]$ have consideration set $Z(b, G) = \{i\}$, but customers with size $b \in [i + 1/4, i + 1/2]$ have consideration set $Z(b, G) = \{i, i + 1\}$. This means a quarter of those customers will substitute with a larger size if the focal size is unavailable. With $t = 7/4$, the substitution window can contain sizes that are two size units away. To see that, notice customers with size $b \in [i - 1/4, i + 1/4]$ have consideration set $Z(b, G) = \{i - 1, i, i + 1\}$, but customers with size $b \in [i + 1/4, i + 1/2]$ have consideration set $Z(b, G) = \{i - 1, i, i + 1, i + 2\}$. We represent the resulting configuration reaction coefficient in (EC.6).

$$f_{t=1/2}^i(C_j) = \begin{cases} 1 & \text{if } i \in C_j, \\ 0 & \text{o.w.} \end{cases}, \quad f_{t=3/4}^i(C_j) = \begin{cases} 1 & \text{if } i \in C_j, \\ \frac{1}{2} & \text{if } i \notin C_j \wedge (i-1 \vee i+1) \in C_j, \\ \frac{1}{4} & \text{if } (i \vee i-1) \notin C_j \wedge i+1 \in C_j, \\ \frac{1}{4} & \text{if } (i \vee i+1) \notin C_j \wedge i-1 \in C_j, \\ 0 & \text{o.w.} \end{cases}, \quad f_{t=7/4}^i(C_j) = \begin{cases} 1 & \text{if } (i \vee i+1 \vee i-1) \in C_j, \\ \frac{1}{2} & \text{if } (i \vee i \pm 1) \notin C_j \wedge (i-2 \wedge i+2) \in C_j, \\ \frac{1}{4} & \text{if } (i \vee i \pm 1 \vee i-2) \notin C_j \wedge i+2 \in C_j, \\ \frac{1}{4} & \text{if } (i \vee i \pm 1 \vee i+2) \notin C_j \wedge i-2 \in C_j, \\ 0 & \text{o.w.,} \end{cases} \quad (\text{EC.6})$$

where “ $\vee$ ” and “ $\wedge$ ” denote the logical OR and AND operators, respectively.

EC.2. Proofs

Proof of Lemma 1 Please refer to Appendix EC.1.

Proof of Lemma 2 For a given $\mathbf{P}$, we have $Unf(\mathbf{P}) = \max_{l,k \in G} \{U^l(\mathbf{P}) - U^k(\mathbf{P})\}$. So, we must have $Unf(\mathbf{P}) \geq U^l(\mathbf{P}) - U^k(\mathbf{P})$ and $Unf(\mathbf{P}) \geq U^k(\mathbf{P}) - U^l(\mathbf{P})$. meaning, $Unf(\mathbf{P}) \geq 0$ for any $\mathbf{P}$.

Take λ_1 and λ_2 such that $0 < \lambda_1 < \lambda_2$. We show the optimal assortment for λ_1 and λ_2 with $\mathbf{P}_1$ and $\mathbf{P}_2$, respectively. We have:

$$\begin{aligned}\Pi(\mathbf{P}_1) &= Rev(\mathbf{P}_1) - \lambda_1 Unf(\mathbf{P}_1) \geq Rev(\mathbf{P}_2) - \lambda_1 Unf(\mathbf{P}_2) \\ &\geq Rev(\mathbf{P}_2) - \lambda_2 Unf(\mathbf{P}_2) = \Pi(\mathbf{P}_2),\end{aligned}$$

where the first inequality holds because $\mathbf{P}_1$ is the optimal solution for $\lambda = \lambda_1$, and the second inequality comes from the fact that $Unf(\mathbf{P}_1) \geq 0$. This proves $\Pi(\mathbf{P}_1) \geq \Pi(\mathbf{P}_2)$. For the revenue and unfairness terms, from the optimality of $\mathbf{P}_1$ and $\mathbf{P}_2$, we write:

$$\begin{aligned}Rev(\mathbf{P}_1) - \lambda_1 Unf(\mathbf{P}_1) &\geq Rev(\mathbf{P}_2) - \lambda_1 Unf(\mathbf{P}_2), \\ Rev(\mathbf{P}_2) - \lambda_2 Unf(\mathbf{P}_2) &\geq Rev(\mathbf{P}_1) - \lambda_2 Unf(\mathbf{P}_1).\end{aligned}$$

Define $\Delta Rev := Rev(\mathbf{P}_1) - Rev(\mathbf{P}_2)$ and $\Delta Unf := Unf(\mathbf{P}_1) - Unf(\mathbf{P}_2)$. We have:

$$\begin{aligned}\Delta Rev &\geq \lambda_1 \Delta Unf, \\ \Delta Rev &\leq \lambda_2 \Delta Unf.\end{aligned}$$

There are four possibilities for the sign of ΔRev and ΔUnf . Suppose they are both strictly negative. Then $|\Delta Rev| \leq \lambda_1 |\Delta Unf|$ contradicts $|\Delta Rev| \geq \lambda_2 |\Delta Unf|$, because $\lambda_2 > \lambda_1$. Also, if just one of them is strictly negative, then multiplying by λ_1 or λ_2 should not change the direction of inequality, meaning this possibility is infeasible. So, the only feasible possibility is $\Delta Rev \geq 0$ and $\Delta Unf \geq 0$. One can easily extend the proof to (AOP-DetFair) by replacing $\mathbf{P}^i$ with S^i , the optimal single assortment for λ_i . $\square$

Proof of Lemma 3 Take two distributions $\mathbf{P}^1$ and $\mathbf{P}^2$ as given. We have the following:

$$\begin{aligned}Rev(\alpha \mathbf{P}^1 + (1 - \alpha) \mathbf{P}^2) &= \sum_{w=1}^{2^n} (\alpha P^1(S^w) + (1 - \alpha) P^2(S^w)) Rev(S^w) \\ &= \alpha \sum_{w=1}^{2^n} P^1(S^w) Rev(S^w) + (1 - \alpha) \sum_{w=1}^{2^n} P^2(S^w) Rev(S^w) \\ &= \alpha Rev(\mathbf{P}^1) + (1 - \alpha) Rev(\mathbf{P}^2).\end{aligned}$$

So, the revenue term is linear in $\mathbf{P}$. For the unfairness term we have:

$$\begin{aligned}Unf(\alpha \mathbf{P}^1 + (1 - \alpha) \mathbf{P}^2) &= \max_{l, k \in G} [U^k(\alpha \mathbf{P}^1 + (1 - \alpha) \mathbf{P}^2) - U^l(\alpha \mathbf{P}^1 + (1 - \alpha) \mathbf{P}^2)] \\ &= \max_{l, k \in G} [\alpha U^k(\mathbf{P}^1) + (1 - \alpha) U^k(\mathbf{P}^2) - \alpha U^l(\mathbf{P}^1) - (1 - \alpha) U^l(\mathbf{P}^2)] \\ &\leq \alpha \max_{l, k \in G} [U^k(\mathbf{P}^1) - U^l(\mathbf{P}^1)] + (1 - \alpha) \max_{l, k \in G} [U^k(\mathbf{P}^2) - U^l(\mathbf{P}^2)],\end{aligned}$$

which proves the convexity of $Unf(\mathbf{P})$ in $\mathbf{P}$.

As we mentioned, we define $Fair(\mathbf{P})$ as the complement of $Unf(\mathbf{P})$. Therefore, $Fair(\cdot)$ is concave in $\mathbf{P}$. For every $\lambda \geq 0$, we define $\mathbf{P}(\lambda)$ as the optimal solution for the given λ . Take any two values $\lambda_1 < \lambda_2$ where $(Rev(\mathbf{P}(\lambda_1)), Fair(\mathbf{P}(\lambda_1)))$ and $(Rev(\mathbf{P}(\lambda_2)), Fair(\mathbf{P}(\lambda_2)))$ are distinct. For any $\lambda \in (\lambda_1, \lambda_2)$ with a $(Rev(\mathbf{P}(\lambda)), Fair(\mathbf{P}(\lambda)))$ distinct from the past two pairs, Lemma 2 implies that

$Rev(\mathbf{P}(\lambda_2)) < Rev(\mathbf{P}(\lambda)) < Rev(\mathbf{P}(\lambda_1))$ and $Fair(\mathbf{P}(\lambda_2)) > Fair(\mathbf{P}(\lambda)) > Fair(\mathbf{P}(\lambda_1))$. We find the unique α for which $Rev(\mathbf{P}(\lambda)) = \alpha Rev(\mathbf{P}(\lambda_1)) + (1 - \alpha) Rev(\mathbf{P}(\lambda_2))$. Suppose $Fair(\mathbf{P}(\lambda)) < \alpha Fair(\mathbf{P}(\lambda_1)) + (1 - \alpha) Fair(\mathbf{P}(\lambda_2))$. From the linearity of $Rev(\mathbf{P})$ and concavity of $Fair(\mathbf{P})$ in $\mathbf{P}$, we know we can use $\mathbf{P}' = \alpha \mathbf{P}(\lambda_1) + (1 - \alpha) \mathbf{P}(\lambda_2)$ and have the same revenue as $Rev(\mathbf{P}(\lambda))$, but a higher fairness. This contradicts the optimality of $\mathbf{P}(\lambda)$. Therefore, the revenue-fairness frontier is concave with respect to the origin. $\square$

Proof of Theorem 1 Take $(\hat{\mathbf{x}}, \hat{\psi})$ as the optimal solution to (AOP-Selection). We construct $\hat{\mathbf{P}}$ as instructed in (12). First, we have the following for every option $j \in N \cup \{0\}$:

$$\Pr(\text{Select } j) = \sum_{k=0}^n \hat{P}(S_k) \Pr(j|S_k) = \sum_{k=j}^n \left(\frac{\hat{x}_k}{v_k} - \frac{\hat{x}_{k+1}}{v_{k+1}} \right) \left[v_0 + \sum_{m \in S_k} v_m \right] \frac{v_j}{v_0 + \sum_{m \in S_k} v_m} = \hat{x}_j.$$

Next, note that:

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}(S) &= \sum_{j=0}^n \hat{P}(S_j) = \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{m \in S_j} v_m \right] = v_0 \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) + \sum_{j=1}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \sum_{m=1}^j v_m \\ &= \hat{x}_0 + \sum_{m=1}^n v_m \sum_{j=m}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) = \hat{x}_0 + \sum_{m=1}^n v_m \frac{\hat{x}_m}{v_m} = 1. \end{aligned}$$

As for the unavailability terms, we have:

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}(S) U^i(s) &= \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{m \in S_j} v_m \right] \left[\frac{v_0 + \sum_{m \in S_j} v_m (1 - f^i(C_m))}{v_0 + \sum_{m \in S_j} v_m} \right] \\ &= v_0 \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) + \sum_{j=1}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \sum_{m=1}^j v_m (1 - f^i(C_m)) \\ &= \hat{x}_0 + \sum_{m=1}^n v_m (1 - f^i(C_m)) \sum_{j=m}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) = \hat{x}_0 + \sum_{m=1}^n \hat{x}_m (1 - f^i(C_m)). \end{aligned}$$

Therefore, for each unavailability pair, we have:

$$\sum_{S \subseteq N} \hat{P}(S) [U^k(S) - U^l(S)] = \sum_{j \in N} \hat{x}_j (1 - f^k(C_j)) - \sum_{j \in N} \hat{x}_j (1 - f^l(C_j)) = \sum_{j \in N} \hat{x}_j (f^l(C_j) - f^k(C_j)).$$

Therefore, by utilizing $\hat{\mathbf{P}}$, the two sets of unfairness constraints in (AOP-Assortment) and (AOP-Selection) have the same upper bound. Similar to the unfairness terms, we can write the revenue term:

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}(S) Rev(S) &= \sum_{j=0}^n \hat{P}(S_j) Rev(S_j) = \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{m \in S_j} v_m \right] \frac{\sum_{m \in S_j} v_m \tilde{r}_m(C_m)}{v_0 + \sum_{m \in S_j} v_m} \\ &= \sum_{j=1}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \sum_{m=1}^j v_m \tilde{r}_m(C_m) = \sum_{m=1}^n v_m \tilde{r}_m(C_m) \sum_{j=m}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) = \sum_{m \in N} \hat{x}_m \tilde{r}_m(C_m). \end{aligned}$$

As such, $(\hat{\mathbf{P}}, \hat{\psi})$ is both feasible in (AOP-Assortment) and brings about the same objective value as $(\hat{\mathbf{x}}, \hat{\psi})$ does in (AOP-Selection). As such, we have that $\Pi^{\text{Assortment-Based}} \geq \Pi^{\text{Selection-Based}}$.

Suppose $(\mathbf{P}', \psi')$ is the optimal solution for (AOP-Assortment). We build a suggested solution for (AOP-Selection) by setting $x'_j := \sum_{S \subseteq [n]} P'(S) \Pr(j|S)$. It is easy to observe that $\sum_{j \in [n]} x'_j + x'_0 = 1$ and $x'_j/v_j \leq x'_0/v_0$. Now, we re-write the unfairness constraints with $\mathbf{x}'$:

$$\begin{aligned} \sum_{j \in N} x'_j (f^k(C_j) - f^l(C_j)) &= \sum_{j \in N} \left(\sum_{S \subseteq N} P'(S) \Pr(j|S) \right) [f^k(C_j) - f^l(C_j)] \\ &= \sum_{S \subseteq N} P'(S) \sum_{j \in N} \Pr(j|S) [f^k(C_j) - f^l(C_j)] \\ &= \sum_{S \subseteq N} P'(S) \sum_{j \in S} \frac{v_j}{v_0 + \sum_{m \in S} v_m} [f^k(C_j) - f^l(C_j)] \\ &= \sum_{S \subseteq N} P'(S) \left[\frac{1 + \sum_{j \in S} v_j (1 - f^l(C_j))}{v_0 + \sum_{m \in S} v_m} - \frac{1 + \sum_{j \in S} v_j (1 - f^k(C_j))}{v_0 + \sum_{m \in S} v_m} \right] \\ &= \sum_{S \subseteq N} P'(S) [U^l(S) - U^k(S)]. \end{aligned}$$

Therefore, by utilizing $\mathbf{x}'$, the two sets of unfairness constraints in (AOP-Assortment) and (AOP-Selection) have the same upper bound. Similar to the unfairness terms, we can write the revenue term:

$$\begin{aligned} \sum_{j \in N} \tilde{r}_j(C_j) x'_j &= \sum_{j \in N} \left(\sum_{S \subseteq N} P'(S) \Pr(j|S) \right) \tilde{r}_j(C_j) = \sum_{S \subseteq N} P'(S) \sum_{j \in N} \Pr(j|S) \tilde{r}_j(C_j) \\ &= \sum_{S \subseteq N} P'(S) \sum_{j \in S} \frac{v_j \tilde{r}_j(C_j)}{v_0 + \sum_{m \in S} v_m} = \sum_{S \subseteq N} P'(S) \text{Rev}(S). \end{aligned}$$

Therefore, $(\mathbf{x}', \psi')$ is both feasible in (AOP-Selection) and brings about the same objective value as $(\mathbf{P}', \beta')$ does in (AOP-Assortment). As such, we have that $\Pi^{\text{Selection-Based}} \geq \Pi^{\text{Assortment-Based}}$.

Combining the two directions together, we proved that: (i) The optimal values of (AOP-Assortment) and (AOP-Selection) are equal. (ii) $\hat{\mathbf{P}}$ provides an optimal solution for (AOP-Assortment). (iii) Expected revenue and unfairness are identical in the two formulations. $\square$

Proof of Theorem 2 The proof steps are exactly the same as the ones for Theorem 1. The only difference is that now we have a distinct set of display probabilities for each size group. Also, for the unavailability terms, we have $u^i(X^i) = x_0^i + \sum_{j \in N} x_j^i (1 - f^i(C_j))$. As such, for each unfairness pair we have:

$$\begin{aligned} u^k(X^i) - u^l(X^k) &= (x_0^k + \sum_{j \in N} x_j^k (1 - f^k(C_j))) - (x_0^l + \sum_{j \in N} x_j^l (1 - f^l(C_j))) \\ &= (x_0^k + \sum_{j \in N} x_j^k) - (x_0^l + \sum_{j \in N} x_j^l) + \sum_{j \in N} x_j^l f^l(C_j) - \sum_{j \in N} x_j^k f^k(C_j) = \sum_{j \in N} x_j^l f^l(C_j) - \sum_{j \in N} x_j^k f^k(C_j), \end{aligned}$$

where the last equality comes from the constraint that $x_0^i + \sum_{j \in N} x_j^i = 1$ for every size group $i \in G$. $\square$

Proof of Theorem 3 We build $\hat{P}_1$ based on $\hat{x}$ similar to the baseline model. We define the mapping σ so that $\hat{x}_{\sigma(1)}/v_{\sigma(1)} \geq \hat{x}_{\sigma(2)}/v_{\sigma(2)} \geq \dots \geq \hat{x}_{\sigma(n)}/v_{\sigma(n)}$. Also, for ease of notation, we define $\sigma(0)$ as the outside option and simply show it with 0, so $\hat{x}_0/v_0 \geq \hat{x}_{\sigma(1)}/v_{\sigma(1)}$. Based on that, nested sets $S_m := \{\sigma(1), \dots, \sigma(m)\}$ with $S_0 = \emptyset$ are defined. Setting $\hat{P}_1(S) = 0$ for any assortment $S \notin \{S_1, \dots, S_n\}$ and $\hat{x}_{n+1} = 0$ we define:

$$\hat{P}_1(S_j) := \frac{\left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}}\right) \left[v_0 + \sum_{i \in S_j} v_i\right]}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} \quad \text{for } j = 0, 1, \dots, n.$$

We build $\hat{P}_{2,(j)}$'s based on $\hat{Q}$ in the following way. We define the mapping σ_j so that $\hat{q}_{(j),\sigma_j(1)}/v_{\sigma_j(1)} \geq \hat{q}_{j,\sigma_j(2)}/v_{\sigma_j(2)} \geq \dots \geq \hat{q}_{j,\sigma_j(n)}/v_{\sigma_j(n)}$. Also, for ease of notation, we define $\sigma_j(0)$ as the outside option and simply show it with 0, so $\hat{q}_{j,0}/v_{j,0} \geq \hat{q}_{j,\sigma_j(1)}/v_{j,\sigma_j(1)}$. Based on that, nested sets $S_{(j),m} := \{\sigma_j(1), \dots, \sigma_j(m)\}$ with $S_{j,0} = \emptyset$ are defined. Setting $\hat{P}_{2,(j)}(S) = 0$ for any assortment $S \notin \{S_{(j),1}, \dots, S_{(j),n}\}$ and $\hat{q}_{j,\sigma_j(n+1)} = 0$, we define:

$$\hat{P}_{2,(j)}(S_{(j),m}) := \frac{\left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}}\right) \left[v_0 + \sum_{t \in S_{(j),m}} v_t\right]}{\hat{q}_{j,0} + \sum_{k \in N} \hat{q}_{j,\sigma_j(k)}} \quad \text{for } m = 0, 1, \dots, n. \quad (\text{EC.7})$$

We begin the prove by showing $(\hat{P}_1, \hat{P}_{2,(1)}, \dots, \hat{P}_{2,(n)})$ and $\hat{\phi}$ constitute a feasible solution for (RBP-Assortment). First, observe that

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}_1(S) &= \frac{1}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} \sum_{j=0}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}}\right) \left[v_0 + \sum_{i \in S_j} v_i\right] \\ &= \frac{v_0}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} \sum_{j=0}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}}\right) + \frac{1}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} \sum_{j=1}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}}\right) \sum_{t=1}^j v_{\sigma(t)} \\ &= \frac{\hat{x}_0}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} + \frac{1}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} \sum_{i=1}^n v_{\sigma(i)} \sum_{j=t}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}}\right) \\ &= \frac{\hat{x}_0}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} + \frac{1}{\hat{x}_0 + \sum_{m \in N} \hat{x}_{\sigma(m)}} \sum_{i=1}^n v_{\sigma(i)} \frac{\hat{x}_{\sigma(i)}}{v_{\sigma(i)}} = 1, \end{aligned}$$

and

$$\begin{aligned}
\sum_{S \subseteq N} \hat{P}_{2,(j)}(S) &= \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{m=0}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \left[v_0 + \sum_{t \in S_{(j),m}} v_t \right] \\
&= \frac{v_0}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{m=0}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \\
&\quad + \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{m=0}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \sum_{t=1}^m v_{\sigma_j(t)} \\
&= \frac{\hat{q}_{j,0}}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} + \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{t=1}^n v_{\sigma_j(t)} \sum_{m=t}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \\
&= \frac{\hat{q}_{j,0}}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} + \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{t=1}^n v_{\sigma_j(t)} \frac{\hat{q}_{j,\sigma_j(t)}}{v_{\sigma_j(t)}} = 1.
\end{aligned}$$

Also, because $\hat{q}_{j,j} = 0$, and because the assortments built through (EC.7) follow a nested structure, $\hat{P}_{2,(j)}(S') = 0$ if $j \in S'$. For the next step in the proof, we establish the equivalence of the first- and second-stage expected numbers of visits per customer for a sample option j between the two models. To see that, we begin with the first stage:

$$\begin{aligned}
\text{Expected number of visits for option } j \\ \text{per-customer at the first stage} &= \sum_{S \subseteq N} \hat{P}_1(S) \Pr(j|S) \\
&= \sum_{m=0}^n \frac{\left(\frac{\hat{x}_{\sigma(m)}}{v_{\sigma(m)}} - \frac{\hat{x}_{\sigma(m+1)}}{v_{\sigma(m+1)}} \right) [v_0 + \sum_{i \in S_m} v_i]}{\hat{x}_0 + \sum_{t=1}^n \hat{x}_{\sigma(t)}} \frac{v_j}{v_0 + \sum_{i \in S_m} v_i} \mathbf{1}(j \in S_m) \\
&= \sum_{m=\sigma^{-1}(j)}^n \frac{\left(\frac{\hat{x}_{\sigma(m)}}{v_{\sigma(m)}} - \frac{\hat{x}_{\sigma(m+1)}}{v_{\sigma(m+1)}} \right) [v_0 + \sum_{i \in S_m} v_i]}{\hat{x}_0 + \sum_{t=1}^n \hat{x}_{\sigma(t)}} \frac{v_j}{v_0 + \sum_{i \in S_m} v_i} \\
&= \frac{v_j}{\hat{x}_0 + \sum_{t \in N} \hat{x}_{\sigma(t)}} \frac{\hat{x}_{\sigma(\sigma^{-1}(j))}}{v_{\sigma(\sigma^{-1}(j))}} = \hat{x}_j,
\end{aligned} \tag{EC.8}$$

where $\sigma^{-1}(j)$ is the rank of option j in the mapping σ . Due to the nested structure of this mapping, for any $m \geq \sigma^{-1}(j)$, we have $j \in S_m$. We now move to establishing a similar result for the second stage. We first need to establish an intermediary result:

$$\begin{aligned}
\sum_{S' \subseteq N} \hat{P}_{2,(e)}(S') \Pr(j|S') &= \sum_{m=0}^n \frac{\left(\frac{\hat{q}_{e,\sigma_e(m)}}{v_{\sigma_e(m)}} - \frac{\hat{q}_{e,\sigma_e(m+1)}}{v_{\sigma_e(m+1)}} \right) [v_0 + \sum_{t \in S_{(e),m}} v_t]}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} \frac{v_j}{v_0 + \sum_{t \in S_{(e),m}} v_t} \mathbf{1}(j \in S_{(e),m}) \\
&= \sum_{m=\sigma_e^{-1}(j)}^n \frac{\left(\frac{\hat{q}_{e,\sigma_e(m)}}{v_{\sigma_e(m)}} - \frac{\hat{q}_{e,\sigma_e(m+1)}}{v_{\sigma_e(m+1)}} \right) [v_0 + \sum_{t \in S_{(e),m}} v_t]}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} \frac{v_j}{v_0 + \sum_{t \in S_{(e),m}} v_t} \\
&= \frac{v_j}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} \frac{\hat{q}_{e,\sigma_e(\sigma_e^{-1}(j))}}{v_{\sigma_e(\sigma_e^{-1}(j))}} = \frac{\hat{q}_{e,j}}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} = \frac{\hat{q}_{e,j}}{\delta \hat{x}_e}.
\end{aligned} \tag{EC.9}$$

The definition of $\sigma_e^{-1}(j)$ and the reason behind the second equation are similar to that of (EC.8). Now, with the use of (EC.8) and (EC.9), we develop the equivalence for the second stage.

$$\begin{aligned} \text{Expected number of visits for option } j \text{ per-customer at second stage} &= \sum_{e \in N} \sum_{S' \subseteq N} \hat{P}_1(S) \Pr(e|S) \delta(1 - f^i(C_e)) \sum_{S' \subseteq N} \hat{P}_{2,(e)}(S') \Pr(j|S') \\ &= \sum_{e \in N} \hat{x}_e \delta(1 - f^i(C_e)) \frac{\hat{q}_{e,j}}{\delta \hat{x}_e} = \sum_{e \in N} \hat{q}_{e,j} (1 - f^i(C_e)) = \hat{y}_j^i. \end{aligned} \quad (\text{EC.10})$$

Now, combining (EC.8) and (EC.10), we have:

$$\text{EV}(j, i | \hat{\mathbf{P}}_1, \hat{\mathbf{P}}_{2,(1)}, \dots, \hat{\mathbf{P}}_{2,(n)}) = \hat{x}_j + \hat{y}_j^i \quad \forall j = 0, 1, \dots, n, i \in G.$$

Using this, it is straightforward to show the equivalence of the revenue and unavailability terms:

$$\begin{aligned} E_{\hat{\mathbf{P}}_1} \sum_{j \in N} E_{\hat{\mathbf{P}}_{2,(j)}} \text{Rev}^i(S_1, S_2) &= \sum_{i \in G} \xi^i \sum_{j \in N} (\hat{x}_j + \hat{y}_j^i) f^i(C_j) r_j, \\ E_{\hat{\mathbf{P}}_1} \sum_{j \in N} E_{\hat{\mathbf{P}}_{2,(j)}} \text{U}^i(S_1, S_2) &= (\hat{x}_0 + \hat{y}_0^i) + \sum_{j \in N} (\hat{x}_j + \hat{y}_j^i) (1 - f^i(C_j)). \end{aligned}$$

Therefore, $(\hat{\mathbf{P}}_1, \hat{\mathbf{P}}_{2,(1)}, \dots, \hat{\mathbf{P}}_{2,(n)})$ and $\hat{\psi}$ constitute a feasible solution for (RBP-Assortment), with the same revenue and level of unfairness.

Now, we focus on the reverse direction. Suppose $(\mathbf{P}'_1, \mathbf{P}'_{2,(1)}, \dots, \mathbf{P}'_{2,(n)})$ and ψ' is the optimal solution for (RBP-Assortment). We build a suggested solution for (RBP-Selection) in the following way:

$$\begin{aligned} x'_j &:= \sum_{S \subseteq N} P'_1(S) \Pr(j|S), \\ y_j^{i'} &:= \sum_{e \in N} \sum_{S' \subseteq N} P'_1(S) \Pr(e|S) \delta(1 - f^i(C_e)) \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S'), \\ q'_{e,j} &:= \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S'). \end{aligned}$$

It is easy to show that $x'_j/v_j \leq x'_0/v_0$ and $q'_{j,m}/v_m \leq q'_{j,0}/v_0$ for all $j, m \in N$. Also, we know $q'_{j,j} = 0$ because $P'_{2,(j)}(S) = 0$ if $j \in S$. We then observe the following:

$$\begin{aligned} \sum_{j=0}^n x'_j &= \sum_{j=0}^n \sum_{S \subseteq N} P'_1(S) \Pr(j|S) = \sum_{S \subseteq N} P'_1(S) \sum_{j=0}^n \Pr(j|S) = \sum_{S \subseteq N} P'_1(S) = 1; \\ \sum_{j=0}^n q'_{e,j} &= \sum_{j=0}^n \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') = \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \sum_{j=0}^n \Pr(j|S') = \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') = \delta x'_e; \\ \sum_{e \in N} q'_{e,j} (1 - f^i(C_e)) &= \sum_{e \in N} \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') (1 - f^i(C_e)) = \sum_{e \in N} \delta x'_e (1 - f^i(C_e)) \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') \\ &= \sum_{e \in N} \sum_{S' \subseteq N} P'_1(S) \Pr(e|S) \delta(1 - f^i(C_e)) \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') = y_j^{i'}. \end{aligned}$$

Also, we have that

$$x'_j + y_j^{i'} = \text{EV}(j, i | \mathbf{P}'_1, \mathbf{P}'_{2,(1)}, \dots, \mathbf{P}'_{2,(n)}) \quad \forall j = 0, 1, \dots, n, i \in G.$$

It is straightforward to show the equivalence of the revenue and discomfort terms:

$$\begin{aligned} \sum_{i \in G} \xi^i \sum_{j \in N} (x'_j + y_{j'}^{i'}) f^i(C_j) r_j &= \sum_{i \in G} \xi^i \sum_{j \in N} \text{EV}(j, i | P', P'_{-1}, \dots, P'_{-n}) f^i(C_j) r_j, \\ (x'_0 + y_0^{i'}) + \sum_{j \in N} (x'_j + y_{j'}^{i'}) (1 - f^i(C_j)) &= \text{EV}(0, i | P', P'_{-1}, \dots, P'_{-n}) + \sum_{j \in N} \text{EV}(j, i | P', P'_{-1}, \dots, P'_{-n}) (1 - f^i(C_j)). \end{aligned}$$

Therefore, $X', \vec{Y}', \vec{Q}'$ and ψ' constitutes a feasible solution for (RBP-Selection), with the same expected revenue and unfairness terms. Combining the two directions, we see that the transformed optimal solution of each approach is feasible in the other approach, and yields the same revenue and unfairness term. $\square$

Proof of Theorem 4 Please refer to Appendix EC.5.

Proof of Lemma 4 We first prove S^{nb} maximizes the revenue term, $\text{Rev}(S)$, in (AOP-DetFair). If $C_j = G$, then $\hat{r}_j = r_j$. Therefore, we have $\hat{r}_j = r_j$ for all $j \in [n]$. So the best option assortment must maximize $(\sum_{j \in S} v_j r_j) / (1 + \sum_{k \in S} v_k)$, which is the set S^{nb} . Therefore, assortment (S^{nb}) maximizes $\text{Rev}(S)$ in (AOP-DetFair).

Now, we prove (S^{nb}) minimizes the unfairness term in (AOP-DetFair). Because $U^i(S^{nb}) = 0$ for every size group i , it follows that $\text{Unf}(S^{nb}) = 0$, which by Lemma 2 is the lower bound for the unfairness term.

So assortment S^{nb} maximizes $\Pi(S) = \text{Rev}(S) - \lambda \text{Unf}(S)$ because it maximizes the revenue term and minimizes the unfairness term. Without a broken assortment, (S^{nb}) is feasible, and hence it is the optimal assortment. $\square$

Proof of Lemma 5 Because the objective function in (AOP-Selection) is bounded and its feasible space is non-empty, there exists a *basic feasible solution* for it (Chen et al. 2022), which we show with $(\hat{x}, \hat{\psi})$. Suppose $\Pi^* > 0$ (otherwise, the optimal solution is $\hat{P}(\emptyset) = 1$, which is a singleton). If $\Pi^* > 0$, we must have $\hat{x}_j > 0$ for at least one option, meaning $\hat{x}_0 > 0$. Because there are $n + 2$ decision variables in (AOP-Selection) and $(\hat{x}, \hat{\psi})$ is a basic feasible solution, at least $n + 2$ of the constraints must be active. Because $\sum_{j \in N} \hat{x}_j + \hat{x}_0 = 1$, at least $n + 1$ of the following constraints must be active:

$$\{\hat{\psi} \geq \lambda \sum_{j \in N} \hat{x}_j (f^k(C_j) - f^l(C_j)) \forall k, l \in G; \frac{\hat{x}_j}{v_j} \leq \frac{\hat{x}_0}{v_0} \forall j \in N; \hat{x}_j \geq 0 \forall j \in N\}.$$

At most g^2 of the first set of constraints are binding, meaning at least $n + 1 - g^2$ of the last two sets of constraints are binding. Note that the last two sets of constraints cannot be active at the same time for each option. That is, for any $j \in N$, $\hat{x}_j = 0$ and $\hat{x}_j/v_j = \hat{x}_0/v_0$ cannot be satisfied at the same time. Denote the total number of options that do not satisfy $\hat{x}_j = 0$ or $\hat{x}_j/v_j = \hat{x}_0/v_0$ with p . The remaining $n - p$ options bind exactly $n - p$ constraints. Since $n + 1 - g^2$ is a lower bound to the number of binding constraints in the last two sets, we must have: $n - p \geq n + 1 - g^2 \Rightarrow p \leq g^2 - 1$.

Now, we interpret p . The way $\hat{P}$ is constructed, options with the same $\hat{x}_j/v_j$ are grouped together and build a bundle. Suppose $\hat{x}_j/v_j = \hat{x}_{j+1}/v_{j+1} = \dots = \hat{x}_{j+l}/v_{j+l}$. The display probability assignment rule in

(12) imposes that $\hat{P}(S_j) = \hat{P}(S_{j+1}) = \dots = \hat{P}(S_{j+l-1}) = 0$. Therefore, the number of bundles is equivalent to the number of distinct assortments that are displayed with a non-zero probability. The p options with $0 < \hat{x}_j/v_j < \hat{x}_0/v_0$ can create at most p bundles. The remaining $n - p$ options fall in one of the two bundles: bundle of options with $\hat{x}_j/v_j = \hat{x}_0/v_0$ or $\hat{x}_j = 0$. However, we can disregard options with $\hat{x}_j = 0$ because they do not create an assortment. Therefore, the number of assortments that are displayed at random with $\hat{P}$ is at most $p + 1$ which is bounded by g^2 . Note that if $\hat{\psi} > 0$ in the optimal solution (meaning unfairness is not fully eradicated in the optimal solution), this limit can be further pushed down to $g^2/4$ for even g 's and $g(g-1)/4$ for odd g 's. $\square$

EC.3. Choice Model Subject to Option Substitution

Unlike physical stores, the online retailer is free to offer any assortment to customers who return to the category page. With the same logic, the retailer can also utilize randomization. Nevertheless, it is safe to assume that the customer will not select option j (her first choice) if it is present in the second-stage assortment. We introduce $N_{-j} := N \setminus \{j\}$ to denote the set of options from which the retailer may choose assortments to display to the returning customer.

As the customer has not yet visited the webpage of other options, we assume her valuation of options in N_{-j} has not changed. So, faced with $S \subseteq N_{-j}$, she will choose option e with probability $v_e/(1 + \sum_{m \in S} v_m)$. After visiting option e 's webpage, the size substitution behavior is similar to that developed in (2). As a reminder, the retailer chooses (i) a first-stage randomized assortment distribution P_1 , and (ii) second-stage randomized assortment distributions $\{P_{2,(1)}, \dots, P_{2,(n)}\}$, where $P_{2,(j)}$ is shown only to customers who look for other options after they did not purchase option j . A modeling complexity here is that, with the possibility of revisits, the ‘probability’ of a customer visiting an option depends on the order of visits and differs between customers leaving at one stage and those opting for a revisit. Therefore, given the possibility of revisits, we focus instead on the number of times an option is visited by each customer who enters the website. We define the ‘expected number of visits per-customer’ (EV) in size group i in the following way:

$$\begin{aligned} \text{EV}(j, i | P_1, P_{2,(1)}, \dots, P_{2,(n)}) &= \underbrace{\sum_{S_1 \subseteq N} P_1(S_1) \Pr(j | S_1)}_{\text{Expected number of visits per-customer at the first stage}} \\ &+ \underbrace{\sum_{e \in N} \sum_{S_1 \subseteq N} P_1(S_1) \Pr(e | S_1) \delta(1 - f^i(C_e)) \sum_{S_2 \subseteq N} P_{2,(e)}(S_2) \Pr(j | S_2)}_{\text{Expected number of visits per-customer at the second stage}}. \end{aligned} \quad (\text{EC.11})$$

where $\Pr(j | S)$ follows the MNL structure defined in (1). The term for the first stage is straightforward, so we focus on the term for the second stage. Suppose a customer in size group i visited option e 's webpage at the first stage. This happens to a fraction $\sum_{S_1 \subseteq N} P_1(S_1) \Pr(e | S_1)$ of entering customers. After that, a fraction $1 - f^i(C_e)$ of customers in size group i leave option e 's webpage, and a fraction δ of them opt to revisit.

These customers are being displayed assortment distribution $P_{2,(e)}$, so fraction $\sum_{S_2 \subseteq N} P_{2,(e)}(S_2) \Pr(j|S_2)$ of them choose option j as their second choice.

A fraction $f^i(C_j)$ of customers in size group i buy an SKU from option j after visiting its webpage, whether at the first or second stage. With this in mind, we write the revenue and unavailability in the revisit approach:

$$\begin{aligned} E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} \text{Rev}^i(S_1, S_2) &:= \sum_{j \in N} \text{EV}(j, i | P, P_{-1}, \dots, P_{-n}) f^i(C_j) r_j, \\ E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} U^i(S_1, S_2) &:= \text{EV}(0, i | P, P_{-1}, \dots, P_{-n}) + \sum_{j \in N} \text{EV}(j, i | P, P_{-1}, \dots, P_{-n}) (1 - f^i(C_j)). \end{aligned} \quad (\text{EC.12})$$

One can derive the closed-form solution for $\text{Rev}^i(S_1, S_2)$ and $U^i(S_1, S_2)$ by setting $P_1(S_1) = 1$ and $P_{2,(e)}(S_2) = 1$ for all options $e \in N$.

EC.4. Optimal Solution for the Hybrid Model

We apply the selection-based transformation to find the optimal assortment. Because the retailer can now tailor both the first and second stage assortments to a customer's size, we introduce the first-stage selection vectors $\vec{X} := \{X^1, \dots, X^g\}$, where $X^i := \{x_1^i, \dots, x_j^i, \dots, x_n^i\}$, for the first stage, and the set of second-stage selection vectors $\vec{Y} := \{Y^1, \dots, Y^g\}$, where $Y^i := \{y_1^i, \dots, y_n^i\}$ for each $i \in G$. We use index 0 to denote the no-purchase choice in both stages. We introduce a set of vectors of auxiliary variables $\mathcal{Q} := \{\vec{Q}^1, \dots, \vec{Q}^g\}$, where $\vec{Q}^i := \{Q_1^i, \dots, Q_n^i\}$ and $Q_j^i := \{q_{j,0}^i, q_{j,1}^i, \dots, q_{j,n}^i\}$ for each $j \in N$ and $i \in G$.

$$\begin{aligned} \Pi_{\text{Hybrid}}^{\text{Selection-Based}} &:= \max_{\psi, \mathbf{x}, \mathbf{Y}, \mathbf{u}, \mathbf{Q}} \sum_{i \in G} \xi^i \sum_{j \in N} (x_j^i + y_j^i) f^i(C_j) r_j - \psi \\ \text{s.t. } \quad &\psi \geq \lambda[u^l - u^k] \quad \forall k, l \in G^2, \\ &u^i = (x_0^i + y_0^i) + \sum_{j \in N} (x_j^i + y_j^i) (1 - f^i(C_j)) \quad \forall i \in G, \\ &\frac{x_j^i}{v_j} \leq \frac{x_0^i}{v_0} \quad \forall j \in N, i \in G, \quad \frac{q_{j,m}^i}{v_m} \leq \frac{q_{j,0}^i}{v_0}, \quad j \in N, m \in N, i \in G, \\ &\sum_{j \in N} x_j^i + x_0^i = 1 \quad \forall i \in G, \quad \sum_{m \in N} q_{j,m}^i + q_{j,0}^i = \delta x_j^i, \quad \forall j \in N, i \in G, \\ &y_m^i = \sum_{j \in N} q_{j,m}^i (1 - f^i(C_j)) \quad i \in G, m \in N \cup \{0\}; \\ &q_{m,j}^i \geq 0 \quad \forall m \in N, j \in N \cup \{0\}, \quad q_{j,j}^i = 0 \quad \forall j \in N, \quad x_j^i \geq 0 \quad \forall j \in N \cup \{0\}, i \in G, \end{aligned} \quad (\text{HP-Selection})$$

where (HP-Selection) stands for *Hybrid Problem - Selection-Based*. We solve this problem and then run a backward transformation similar to that of the revisit behavior model to obtain allocation $\hat{P}$. We now prove the following result.

THEOREM EC.1. (i): *The optimal value of (HP-Assortment) and (HP-Selection) are equal.* (ii) *Allocation $\hat{\mathcal{P}}$ achieves the optimal solution of (HP-Assortment).* (iii) *Expected revenue and unfairness are identical in the two formulations.*

To save space, we ignore the detailed proof of Theorem EC.1. However, by comparing (HP-Selection) and (RBP-Selection), one can easily see their similarities. The only difference here is that we allow the first-stage decision to be curated for each size group.

EC.5. Mathematical Details of Endogenizing Configuration

The conversion from the optimal solution for (ECP-Selection) to (ECP-Assortment) happens in two stages. First, we re-index $\hat{x}$ so that $\hat{x}_1/v_1 \geq \dots \geq \hat{x}_n/v_n$. Then, we define S_m and S_0 similar to the baseline model, with $\hat{P}(S_m)$ defined the same way as (12). We define $\hat{P}_j := \sum_{m=j}^n \hat{P}(S_m)$, which shows the probability of displaying an assortment with option j in it. We develop Algorithm 1 for the second stage probabilities.

Fundamentally, what Algorithm 1 does is ensure that for each option j , each display configuration H gets the display probability $z_{j,H}$. This is feasible because we can *decouple* option and configuration selection decisions. Since we assumed the customer cannot observe the displayed configuration of option j until she chooses that option, x_j and $z_{j,H}$ are connected only through $\sum_{H \subseteq C_j} z_{j,H} = x_j$. Therefore, we first compute the display probabilities for each option, then implement the configuration selection decisions.

The challenge here is managing the display probability of each configuration. Suppose the optimal solution of the first stage consists of displaying a single assortment with option j in it: $\hat{P}(S_1 = \{1, \dots, j, \dots, m\}) = 1$. However, only for option j , we have $\hat{z}_{j,H_1}/\hat{x}_j = 0.5$ and $\hat{z}_{j,H_2}/\hat{x}_j = 0.5$, so we must ensure that half of the customers clicking on option j must see it in configuration H_1 , and the other half must see it in configuration H_2 . To do so, we conduct *assortment mitosis*: we make a copy (i.e., everything the same) of assortment S_1 , call it assortment S_2 , and set the configuration of option j in S_1 as H_1 , and in S_2 as H_2 . Then, we randomize over S_1 and S_2 with equal probability. First, note that the selection probability of option j before and after mitosis remains unchanged, because the options in S_1 and S_2 are the same. Second, now half of the customers choosing option j see it in configuration H_1 , and the other half must see it in configuration H_2 .

Algorithm 1: Assortment-Configuration-Selection

Data: Items $N = \{1, \dots, n\}$; Initial Assortments $\{S_m\}_{m=0}^n$; Initial Display Probabilities $\{\hat{P}(S_m)\}_{m=0}^n$; Configuration Probabilities $\hat{Z}$

Result: A list of weighted Assortment-Configuration tuples $\mathcal{A} = \{A_k\}$, where $\sum_k P_k = 1$.

// Initialization: Create a list of tuples $\mathcal{A} = (S, \mathbf{C}, P)$

```

1 Initial Assortments  $\mathcal{S} \leftarrow \{S_0, S_1, \dots, S_N\}$ 
2  $\mathcal{A} \leftarrow \emptyset$ 
3 for  $m \leftarrow 1$  to  $n$  do
4    $\mathbf{C}_m \leftarrow \mathbf{0}$  // Initialize with empty configurations for all options
5    $P_m \leftarrow \hat{P}_m$  // Assign option probabilities
6    $\mathcal{A} \leftarrow \mathcal{A} \cup \{(S_m, \mathbf{C}_m, P_m)\}$ 
7 end
// Iterate over options to sequentially assign configurations
8 for  $j \leftarrow 1$  to  $n$  do
9   if  $P_j = 0$  then
10    | Break // Skip the option if target probability is 0
11   end
// Sort  $\mathcal{A}$  to prioritize assortments containing  $j$ 
12    $\mathcal{A}_{\text{has},j} \leftarrow \{A_k \in \mathcal{A} : j \in S_k\}$ 
13    $\mathcal{A}_{\text{no},j} \leftarrow \{A_k \in \mathcal{A} : j \notin S_k\}$ 
14    $\mathcal{A} \leftarrow \mathcal{A}_{\text{has},j}$  followed by  $\mathcal{A}_{\text{no},j}$ 
15    $k \leftarrow 1$  // Current index in the list  $\mathcal{A}$ 
// Iterate over all possible configurations
16   for every  $H \subseteq C_j$  do
17      $p_{\text{remain}} \leftarrow \hat{z}_{j,H}$  // Target marginal probability for  $(j, \text{config } H)$ 
18     while  $p_{\text{remain}} \geq P_k \cdot \Pr(j|S_k)$  do
19       // Option  $j$  in assortment  $A_k$  is assigned to configuration  $H$ 
20        $\mathbf{C}_k[j] \leftarrow H$ 
21        $p_{\text{remain}} \leftarrow p_{\text{remain}} - P_k \cdot \Pr(j|S_k)$ 
22        $k \leftarrow k + 1$ 
23     end
24     if  $0 < p_{\text{remain}} < P_k \cdot \Pr(j|S_k)$  then
25       // Assortment Mitosis: Create a copy of  $A_k$ , insert it after
26       //  $A_k$ , and shift indices
27        $A_{\text{new}} \leftarrow \text{Copy}(A_k)$ 
28        $\mathcal{A} \leftarrow \text{Insert}(A_{\text{new}}, \text{after } k)$  // New assortment is at  $k+1$ 
29       // Recalculate probabilities for  $A_k$  and  $A_{k+1}$ 
30        $P_{k+1} \leftarrow P_k - p_{\text{remain}} / \Pr(j|S_k)$ 
31        $P_k \leftarrow p_{\text{remain}} / \Pr(j|S_k)$ 
32       // Assign configuration  $H$  to  $A_k$ 
33        $\mathbf{C}_k[j] \leftarrow H$ 
34        $k \leftarrow k+1$  // Move to the next (unassigned) assortment
35     end
36   end
37 end

```

We now prove Theorem 4. Suppose $(\hat{x}, \hat{Z}, \hat{\psi})$ is the optimal solution for (ECP-Selection). We first get $\hat{P}$ using (12) and then run Algorithm 1 to get $\mathcal{A}$. We define the list of weighted Assortment-Configuration tuples after j -th iteration with $\mathcal{A}^j$. We have the following lemma.

LEMMA EC.1. *After running Algorithm 1, We have:*

1. $\sum_{(S,C,P) \in \mathcal{A}^n} P(S, C) \Pr(j|S) = \hat{x}_j$ for any option j ;
2. $\sum_{(S,C,P) \in \mathcal{A}^n} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \hat{z}_{j,H}$ for any option-configuration pair (j, H) .

Proof of Lemma EC.1 We prove this lemma using induction. define $\mathcal{B}_{has,m}^j \leftarrow \{A_k \in \mathcal{A}^j : m \in S_k\}$ and $\mathcal{D}_{has,m,H}^j \leftarrow \{A_k \in \mathcal{A}^j : m \in S_k, C_m = H\}$. Before the first iteration, we have

$$\sum_{(S,C,P) \in \mathcal{B}_{has,j}^0} P(S, C) \Pr(j|S) = \sum_{m=j}^n \hat{P}(S_j) \Pr(j|S) = \sum_{m=j}^n \left(\frac{\hat{x}_m}{v_m} - \frac{\hat{x}_{m+1}}{v_{m+1}} \right) \left[v_0 + \sum_{i \in S_m} v_i \right] \frac{v_j}{v_0 + \sum_{i \in S_m} v_i} = \hat{x}_j.$$

Similarly, we have $\sum_{(S,C,P) \in \mathcal{D}_{has,j,0}^0} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = C_j[0]) = \hat{x}_j$, and for any $H \neq 0$ we have $\sum_{(S,C,P) \in \mathcal{D}_{has,j,q}^0} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = 0$.

When running the first iteration, we divide our assortment into two mutually exclusive subsets $\mathcal{B}_{has,1}^0$ and $\mathcal{A}^0 \setminus \mathcal{B}_{has,1}^0$. Assortments in $\mathcal{A}^0 \setminus \mathcal{B}_{has,1}^0$ are not affected by the first iteration. That is, $\mathcal{A}^0 \setminus \mathcal{B}_{has,1}^0 = \mathcal{A}^1 \setminus \mathcal{B}_{has,1}^1$. However, the same cannot be said for assortments in $\mathcal{B}_{has,1}^0$ because of assortment modification and/or assortment mitosis during the first iteration.

Suppose during the first iteration, Algorithm 1 modifies assortment A_k . We know the modification process during the first iteration does not affect the configurations of any option other than option 1. As such, we have that $P_k \Pr(j|S_k) \mathbf{1}(C_k[j] = H)$ remains unchanged. Now, suppose that during the first iteration, Algorithm 1 runs mitosis on assortment A_l and produces a modified A_l and A_{l+1} . Because A_{l+1} is a copy of A_l , we have $\Pr(j|S_l) \mathbf{1}(C_l[j] = H) = \Pr(j|S_{l+1}) \mathbf{1}(C_{l+1}[j] = H)$. Also, we have that $P_l + P_{l+1}$ after mitosis is equal to P_l before mitosis. That means $P_l \Pr(j|S_l) \mathbf{1}(C_l[j] = H) + P_{l+1} \Pr(j|S_{l+1}) \mathbf{1}(C_{l+1}[j] = H)$ after mitosis and $P_l \Pr(j|S_l) \mathbf{1}(C_l[j] = H)$ before mitosis are equal. Therefore, for any configuration H and any option except option 1, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,1}^1} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{B}_{has,1}^0} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$. Using that, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,1}^1} P(S, C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{B}_{has,1}^0} P(S, C) \Pr(j|S)$. Combining the two types of assortments, for any option-configuration pairs (j, H) with $j \neq 1$ we have $\sum_{(S,C,P) \in \mathcal{A}^1} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{A}^0} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$, which in turn means $\sum_{(S,C,P) \in \mathcal{A}^1} P(S, C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{A}^0} P(S, C) \Pr(j|S) = \hat{x}_j$.

We now explain the changes toward option 1 in the first iteration. For this, consider configuration H . Without loss of generality, suppose through modification and/or mitosis, Algorithm (1) assigns configuration H to assortments A_1 to A_m . From the way that p_{remain} is updated in the algorithm, we know that $\sum_{k=1}^m P_k \Pr(j|S_k) = \hat{z}_{j,H}$. Therefore, $\sum_{(S,C,P) \in \mathcal{A}^1} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_1 = H) = \hat{z}_{1,H}$. Also, from the relevant constraint in (ECP-Selection), we know that

$$\sum_{(S,C,P) \in \mathcal{A}^1} P(S, C) \Pr(1|S) = \sum_{H \subseteq C_j} \sum_{(S,C,P) \in \mathcal{A}^1} P(S, C) \Pr(1|S) \cdot \mathbf{1}(C_1 = H) = \sum_{H \subseteq C_j} \hat{z}_{1,H} = \hat{x}_1.$$

Therefore, $\sum_{(S,C,P) \in \mathcal{A}^1} P(S, C) \Pr(j|S) = \hat{x}_j$ for any option j , and $\sum_{(S,C,P) \in \mathcal{A}^1} P(S, C) \Pr(1|S) \cdot \mathbf{1}(C_1 = H) = \hat{z}_{1,H}$ for any configuration H . This concludes the first step of induction.

Suppose $\sum_{(S,C,P) \in \mathcal{A}^k} P(S, C) \Pr(j|S) = \hat{x}_j$ for any option j , and $\sum_{(S,C,P) \in \mathcal{A}^k} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \hat{z}_{j,H}$ for all option-configuration pairs (j, H) where $j \leq k$. We run the $k + 1$ -th iteration. The proof steps are the same as those we followed for the first iteration. We first divide $\mathcal{A}^{k+1}$ into two groups: $\mathcal{B}_{has,k+1}^{k+1}$ and $\mathcal{A}^{k+1} \setminus \mathcal{B}_{has,k+1}^{k+1}$. We know that the algorithm does not change the configuration of any option other than option $k + 1$ during the $k + 1$ -th iteration. Therefore, for any option-configuration pair (j, H) with $j \neq k + 1$, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^{k+1}} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^k} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$. Using that, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^{k+1}} P(S, C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^k} P(S, C) \Pr(j|S)$. Combining the two types of assortments, for any option-configuration pairs (j, H) with $j \neq k + 1$ we have $\sum_{(S,C,P) \in \mathcal{A}^{k+1}} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{A}^k} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$, which in turn means $\sum_{(S,C,P) \in \mathcal{A}^{k+1}} P(S, C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{A}^k} P(S, C) \Pr(j|S) = \hat{x}_j$. Also, from the induction assumption, we see that $\sum_{(S,C,P) \in \mathcal{A}^{k+1}} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{A}^k} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \hat{z}_{j,H}$ for any $j \leq k$. We then follow the similar steps as what we did for option 1 to prove that $\sum_{(S,C,P) \in \mathcal{A}^{k+1}} P(S, C) \Pr(j|S) \cdot \mathbf{1}(C_{k+1} = H) = \hat{z}_{k+1,H}$, so $\sum_{(S,C,P) \in \mathcal{A}^{k+1}} P(S, C) \Pr(k + 1|S) = \hat{x}_{k+1}$ after running $k + 1$ -th iteration, concluding the proof. $\square$

With Lemma EC.1, proving that the optimal solution for (ECP-Selection) can be converted into a feasible solution for (ECP-Assortment) with the same revenue and unfairness terms is straightforward and follows the same steps as in the baseline model. For the reverse relation, suppose P' is the optimal solution for (ECP-Assortment). Define:

$$x'_j = \sum_{P \in P'} \sum_{j \in N} \sum_H P(S, C) \cdot \Pr(j|S) \cdot \mathbf{1}(C_j = H), \quad z'_{j,H} = \sum_{P \in P'} \sum_{j \in N} P(S, C) \cdot \Pr(j|S) \cdot \mathbf{1}(C_j = H).$$

Using these relations, proving the optimal solution for (ECP-Assortment) with the same revenue and unfairness term is feasible in (ECP-Selection) is straightforward. Combining the two directions, the equivalence of the two programs is proven.

EC.6. Model Calibration

In this section, we apply a Difference-in-Difference (DiD) method to estimate the model coefficients using a granular, SKU-level dataset we acquired while collaborating with an industry partner.

EC.6.1. The Dataset

Our industry partner is a major European fast fashion retailer that provided us with access to SKU-level sales data for its online store for a span of two months (late January to late February 2019). We focus on the women's long-sleeve shirts category in this study. On average, 238 distinct options and 970 distinct SKUs were displayed daily on this category's page. The average daily sales from this category is 80, which shows an average sales-per-SKU record of 0.082.

EC.6.2. DiD Analysis and the Structural Model

To estimate the values of p^u and p^l in (2), we adopt the DiD approach in Li et al. (2023). The level of granularity in our data allows us to adopt the DiD method in Li et al. (2023), which does not require complex estimation methods or assumptions on latent variables (such as the methods employed in, e.g., Vulcano et al. (2012) and Musalem et al. (2010)).

The key idea in Li et al. (2023) is to focus on the SKU-level data and to choose the control and treatment groups from different sizes of the same option. This approach eliminates the factors common to different sizes of an option and focuses on the effects caused by differences in size groups. This is particularly relevant to the in-store shoe shopping experience that they study as all customers who visit a store on a given day see the same assortment of shoes. Upon choosing the shoes (brand, design, color, etc.) that they prefer, customers ask for the shoes in their size. So to the point of checking for availability of an option in their size, customers of all sizes have similar experiences. Because our baseline choice model assumes the same, we adopt their method to estimate the parameters of interest.

Suppose on the day t , N_t customers (in all sizes) visit the online store with the assortment (S_t). The expected number of customers who visit the page for option j on that day is $N_t \frac{v_j}{1 + \sum_{k \in S_t} v_k}$, which is independent of the configuration. Sales of this option (i.e., SKUs that belong to it) depend on its configuration. More specifically, the expected sales of size i of option j are:

$$E[\text{Sales}_{jt}^i | S_t, C_t, N_t] = N_t \frac{v_j}{1 + \sum_{k \in S_t} v_k} \mathbf{1}_{(i \in C_{jt})} (\xi^i + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \xi^{i-1} + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \xi^{i+1}), \quad (\text{EC.13})$$

where $C_t := \{C_{1t}, \dots, C_{nt}\}$. In (EC.13), an SKU's sales depend not only on its availability but also on the (un)availability of its neighboring sizes. This helps us to choose the treatment and control groups similar to Li et al. (2023). The following example elaborates on the approach.

Suppose the available size groups are $G = \{XS, S, M, L, XL, XXL\}$, and on day 1, $C_{j1} = G$, i.e., all the sizes of option j are available and so size substitution for this option is not a concern. But on day 2, $C_{j2} = G \setminus \{S\}$, so all sizes for option j are available except size S. We use equation (EC.13) to write the expected sales of sizes M and L over two days in Table EC.1:

	Day 1	Day 2
$E[\text{Sales}_j^M S, C, N]$	$N_1 \frac{v_j}{1 + \sum_{k \in S_1} v_k} \xi^M$	$N_2 \frac{v_j}{1 + \sum_{k \in S_2} v_k} (\xi^M + \xi^S p^u)$
$E[\text{Sales}_j^L S, C, N]$	$N_1 \frac{v_j}{1 + \sum_{k \in S_1} v_k} \xi^L$	$N_2 \frac{v_j}{1 + \sum_{k \in S_2} v_k} \xi^L$

Table EC.1 Expected sales of option j for SKUs S and M according to (EC.13).

Table EC.1 provides the structure of the DiD estimations. We divide the shocks affecting an option's sales into two groups. The first group consists of factors that depend on option assortment on day t (i.e., S_t), and the second group encompasses factors depending on the configuration of option j on that day (i.e., C_{jt}).

Both M and L sizes follow *parallel trends* in the sense that changes in S_t affect both similarly. However, by equation (EC.13), unavailability in size S of option j affects the expected sales of size M but not the sales of size L. In other words, had it not been for the unavailability of size S, sizes M and L of option j would have followed a parallel trend.

Assuming that N_t follows a Poisson distribution (see e.g., Vulcano et al. (2012) and Li et al. (2023)), it must be that $E[\text{Sales}_{jt}^i | S_t, C_t]$ follows a Poisson distribution, i.e., $E[\text{Sales}_{jt}^i | S_t, C_t] \sim \text{Poisson}(\phi_{jt}^i)$. Therefore, using (EC.13), for the days where $j \in S_t$ and $i \in C_{jt}$, we have:

$$\begin{aligned} \log(\phi_{jt}^i) &= \log\{E[\text{Sales}_{jt}^i | S_t, C_t]\} = \log\left(\frac{N_t}{1 + \sum_{k \in S_t} v_k}\right) + \log(v_j) + \log(\xi^i) \\ &\quad + \log\left(1 + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \frac{\xi^{i-1}}{\xi^i} + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \frac{\xi^{i+1}}{\xi^i}\right) \\ &= \alpha_t + \beta_j + \zeta_i + SSE_{jt}^i, \end{aligned} \quad (\text{EC.14})$$

where $SSE_{jt}^i = \log\left(1 + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \xi^{i-1} / \xi^i + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \xi^{i+1} / \xi^i\right)$ captures the size substitution effect. The last equality in (EC.14) holds because the first term on the right-hand side is only a function of t , the second term is only a function of the option j 's perceived value, and the third term is only a function of the fraction of people in size-group i .

As SSE_{jt}^i is nonlinear, we use a linear approximation. The two parameters p^u and p^l are unlikely to be large numbers and so we apply the first-order Taylor expansion of $\log(1 + x)$ to write: $SSE_{jt}^i \cong p^u \mathbf{1}_{(i-1 \notin C_{jt})} \xi^{i-1} / \xi^i + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \xi^{i+1} / \xi^i$. This helps rewrite equation (EC.14) as

$$\log(\phi_{jt}^i) \cong \alpha_t + \beta_j + \zeta_i + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \frac{\xi^{i-1}}{\xi^i} + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \frac{\xi^{i+1}}{\xi^i}. \quad (\text{EC.15})$$

To estimate (EC.15), we define six scenarios. In Scenario 1, we estimate the size substitution effect of unavailability in size XS on size S. To do so, we track four sizes of option j : S, M, L, and XL. We only focus on the days when sizes M, L, and XL are available—size S may or may not be present. Similar to the example in Table EC.1, the sales for size L of option j should not be affected by the absence or presence of size S, as both of its neighbors are available to the customers. This explains why size XL needs to be in the assortment as well. For size M, on the other hand, we expect to observe the size substitution effect from size S to size M on days when size S is absent from the assortment. Therefore, in Scenario 1, we choose size M as our *treatment* and size L as our *control* groups. With the same logic, we introduce the remaining five scenarios in Table EC.2. The first three scenarios occur when a larger size is substituted for an unavailable size, while the last three scenarios occur when a smaller size is substituted.

Based on these scenarios, we create a DiD regression dataset for several categories. Obviously, for identification in each scenario, we focus on options for which there were days when they were available in all four sizes and days when they were unavailable in a size.

Scenario	Out-of-Stock(o)	Treatment(t)	Control(c)	Neighbor(n)
1	XS	S	M	L
2	S	M	L	XL
3	M	L	XL	XXL
4	L	M	S	XS
5	XL	L	M	S
6	XXL	XL	L	M

Table EC.2 Scenarios in the DiD Regressions.

Given the setup of the DiD model, a treatment size cannot be the target of the substitution effect by two neighboring sizes at the same time. So if an option is used for a scenario, we do not use it for another ‘conflicting scenario’ (e.g., for Scenarios 2 and 3 where the treatment and control sets overlap). Overall:

$$\log(\phi_{jt}^i) = \pi + p \times \frac{\xi^o}{\xi^t} \times \text{Treat}_j^{io} \times \text{OOS}_{jt}^o + \kappa \times \text{Treat}_j^{io} + \theta \times \text{OOS}_{jt}^o + \phi \log(\text{price}_{jt}^i) + \alpha_t + \beta_j + \zeta_i + \varepsilon_{jti}. \quad (\text{EC.16})$$

The variable p is the size substitution coefficient to be estimated; Treat_j^{io} is equal to 1 if, in the scenario under discussion, size i is the neighbor of the out-of-stock size ‘ o ’ and 0 otherwise; OOS_{jt}^o is equal to 1 if size ‘ o ’ of option j is out of stock on day t and 0 otherwise; ϕ is the effect of $\log(\text{price}_{jt}^i)$ (we add this term to control for changes in prices during the study period); and π is the constant term. We do not include the inventory level of an SKU as this information is not available to the customers.

Independent Variables	Women Shirt LS		
	All	Sc 1, 2, 3	Sc 4, 5, 6
$\text{Treat}_j^{io} \times \text{OOS}_{jt}^o$	0.2349** (0.1171)	0.0643 (0.2941)	0.2568** (0.1283)
$\log(\text{price})$	-5.3391** (2.1514)	-0.4689 (5.0332)	-5.1724** (2.1535)
Date FE	YES	YES	YES
Option FE	YES	YES	YES
Size FE	YES	YES	YES
Observations	12128	2574	9554
Full Log-Likelihood	3761	666	3044

Table EC.3 Unweighted regression results for ‘Women Shirts Long Sleeve’. * $p < 0.1$, ** $p < 0.05$, * $p < 0.01$.**

We run these regressions in three parts. First, we pool all the samples and assume $p^u \approx p^l$ to obtain a single estimation. To estimate p^u , we run the regression model on samples from Scenarios 1–3 in which size substitution happens with one size larger. The last regression is on samples from Scenarios 4–6 to estimate p^l . Table EC.3 presents the results of the DiD regression on the target category. Due to data limitations,

the results for Scenarios 1–3 are insignificant. Nonetheless, the other two regressions show a significant substitution factor of approximately 0.25 - similar to what was found in Li et al. (2023).

(We removed the details about the regression results for two other product categories, the placebo test, and the specification test from the electronic companion to comply with the page limit requirements of MSOM. They are available upon request.)

EC.6.3. Simulation Setup

We follow a number of steps to characterize the distribution of C_j 's (i.e., available sizes in configurations) from the dataset. First, we create a new coding 'option-date-code' to assign a unique number to each option (and its SKUs) for each day. Next, we observe the availability in different sizes for each option-date pair. We set $z_{jt}^i = 1$ if option j is available (has at least one SKU in the inventory) in size i on day t . If an option is absent from the assortment on a day, we do not define z_{jt}^i for that option-date pair. If option j is not produced in size i , we also set $z_{jt}^i = 0$ for every day that option j is present in the assortment. Finally, we take the average over all option-date pairs; the results are in Table 1. In the simulations, we set the availability probability of an option in each size to follow a Bernoulli distribution with parameter p as in Table 1 and assume the availability of different sizes is independent of each other.

For customer size distribution, we use the estimated Ξ for women's shirts category in Table 1. Table 1 reveals the systematic discrimination that we want to alleviate: While size groups XXL and L have the same number of customers, the unavailability rate for size group XXL is almost twice that of size group L .

The next step is choosing the distribution of preference weights ($\{v_j\}$) and revenues ($\{r_j\}$). While prices are available in our dataset, the preference weights must be inferred from the data. By (EC.14), the option-level fixed effects (β_j 's) in regressions represent $\log(v_j)$. Since we set $v_j = \exp(u_j)$ in Section 3, it is the case that $u_j = \log(v_j) = \beta_j$. Therefore, after performing the weighted DiD regressions, we set $\hat{u}_j = \hat{\beta}_j$, where the hat sign shows the estimation results. Figure EC.2 plots $\hat{u}_j$ against r_j for each option as well as their histograms.

We make multiple observations from Figure EC.2. First, and typical for a fast fashion retailer, prices are distributed according to a unimodal and right-skewed distribution. Second, $\hat{u}_j$'s tend to follow a unimodal and symmetric distribution. Third, a negative correlation exists between r_j 's and $\hat{u}_j$'s.

We use a Gamma distribution with shape = 10 and scale = 5 to obtain price samples. To mimic the price distribution of the focal category, we assume prices cannot be less than 25 or more than 100. As such, if a price is higher than 100 or less than 25, we replace it with a sample from a uniform distribution between 25 and 100. (For a Gamma distribution with shape = 10 and scale = 5, the mean is 50. Considering the mean of the replaced samples is $62.5 = (100 + 25)/2$, the mean of r_j in our simulations is slightly above 50.)

Next, we obtain a_j as samples of a standard Normal distribution and then build $u_j = a_j - \nu r_j$ with $\nu = 0.02$ to represent the slope. Finally, we set $v_j = \exp(u_j)$ to get the preference weights used in the

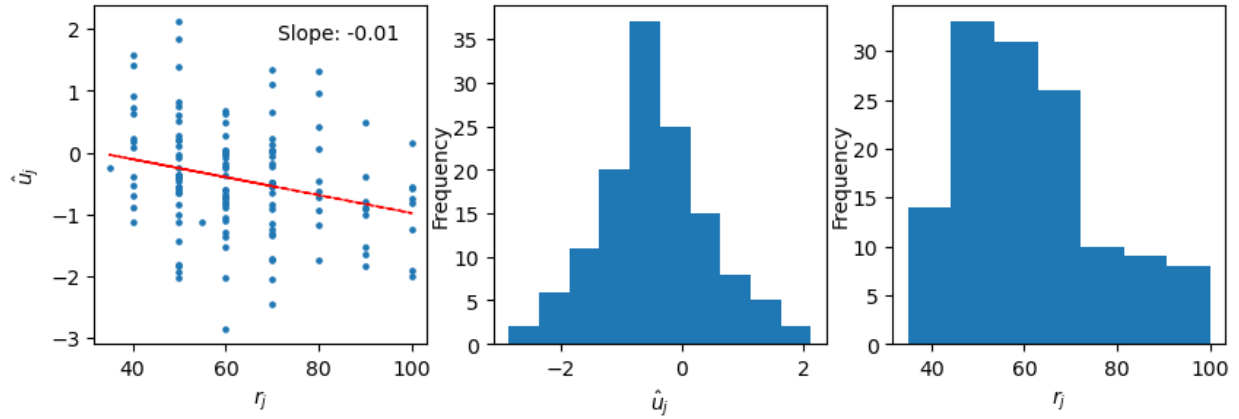


Figure EC.2 $\{r_j\}$ and $\{\hat{u}_j\}$ distribution for women's shirts category.

equations. We do not have access to the number of customers who visited the store on a specific day, so we do not know how many customers left the store without choosing anything. Similar to Sumida et al. (2021), we scale preference weights so that $1/(1 + \sum_{j \in [n]} v_j)$ (the probability that a customer leaves the store without choosing any option if all the options are introduced) is equal to an exogenously set value p_0 .